Neural Computation -- 2007

176 article(s), all issues

1. Mean-driven and fluctuation-driven persistent activity in recurrent networks

RenartAlfonso, Moreno-BoteRubén, WangXiao-Jing, PargaNéstor

https://doi.org/10.5555/1238630.1238631

Spike trains from cortical neurons show a high degree of irregularity, with coefficients of variation (CV) of their interspike interval (ISI) distribution close to or higher than one. It has been s...

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2. (untitled)

(authors unknown)

https://doi.org/10.1162/neco.2007.19.issue-10

No abstract available.

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3. (untitled)

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14. (untitled)

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https://doi.org/10.1162/neco.2007.19.issue-5

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15. The High-Conductance State of Cortical Networks

Arvind Kumar, Sven Schrader, Ad Aertsen, Stefan Rotter

https://doi.org/10.1162/neco.2008.20.1.1 | pp. 1-43

We studied the dynamics of large networks of spiking neurons with conductance-based (nonlinear) synapses and compared them to networks with current-based (linear) synapses. For systems with sparse and inhibition-dominated recurrent connectivity, weak external inputs induced asynchronous irregular firing at low rates. Membrane potentials fluctuated a few millivolts below threshold, and membrane conductances were increased by a factor 2 to 5 with respect to the resting state. This combination of parameters characterizes the ongoing spiking activity typically recorded in the cortex in vivo. Many aspects of the asynchronous irregular state in conductance-based networks could be sufficiently well characterized with a simple numerical mean field approach. In particular, it correctly predicted an intriguing property of conductance-based networks that does not appear to be shared by current-based models: they exhibit states of low-rate asynchronous irregular activity that persist for some period of time even in the absence of external inputs and without cortical pacemakers. Simulations of larger networks (up to 350,000 neurons) demonstrated that the survival time of self-sustained activity increases exponentially with network size.

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16. Irregular Firing of Isolated Cortical Interneurons in Vitro Driven by Intrinsic Stochastic Mechanisms

Bernhard Englitz, Klaus M. Stiefel, Terrence J. Sejnowski

https://doi.org/10.1162/neco.2008.20.1.44 | pp. 44-64

Pharmacologically isolated GABAergic irregular spiking and stuttering interneurons in the mouse visual cortex display highly irregular spike times, with high coefficients of variation approximately 0.9-3, in response to a depolarizing, constant current input. This is in marked contrast to cortical pyramidal cells, which spike quite regularly in response to the same current injection. We applied time-series analysis methods to show that the irregular behavior of the interneurons was not a consequence of low-dimensional, deterministic processes. These methods were also applied to the Hindmarsh and Rose neuronal model to confirm that the methods are adequate for the types of data under investigation. This result has important consequences for the origin of fluctuations observed in the cortex in vivo.

17. A Very Simple Spiking Neuron Model That Allows for Modeling of Large, Complex Systems

Jeffrey J. Lovelace, Krzysztof J. Cios

https://doi.org/10.1162/neco.2008.20.1.65 | pp. 65-90

This letter introduces a biologically inspired very simple spiking neuron model. The model retains only crucial aspects of biological neurons: a network of time-delayed weighted connections to other neurons, a threshold-based generation of action potentials, action potential frequency proportional to stimulus intensity, and interneuron communication that occurs with time-varying potentials that last longer than the associated action potentials. The key difference between this model and existing spiking neuron models is its great simplicity: it is basically a collection of linear and discontinuous functions with no differential equations to solve. The model's ability to operate in a complex network was tested by using it as a basis of a network implementing a hypothetical echolocation system. The system consists of an emitter and two receivers. The outputs of the receivers are connected to a network of spiking neurons (using the proposed model) to form a detection grid that acts as a map of object locations in space. The network uses differences in the arrival times of the signals to determine the azimuthal angle of the source and time of flight to calculate the distance. The activation patterns observed indicate that for a network of spiking neurons, which uses only time delays to determine source locations, the spatial discrimination varies with the number and relative spacing of objects. These results are similar to those observed in animals that use echolocation.

18. Bayesian Spiking Neurons I: Inference

Sophie Denève

https://doi.org/10.1162/neco.2008.20.1.91 | pp. 91-117

We show that the dynamics of spiking neurons can be interpreted as a form of Bayesian inference in time. Neurons that optimally integrate evidence about events in the external world exhibit properties similar to leaky integrate-and-fire neurons with spike-dependent adaptation and maximally respond to fluctuations of their input. Spikes signal the occurrence of new information-what cannot be predicted from the past activity. As a result, firing statistics are close to Poisson, albeit providing a deterministic representation of probabilities.

19. Bayesian Spiking Neurons II: Learning

Sophie Denève

https://doi.org/10.1162/neco.2008.20.1.118 | pp. 118-145

In the companion letter in this issue ("Bayesian Spiking Neurons I: Inference"), we showed that the dynamics of spiking neurons can be interpreted as a form of Bayesian integration, accumulating evidence over time about events in the external world or the body. We proceed to develop a theory of Bayesian learning in spiking neural networks, where the neurons learn to recognize temporal dynamics of their synaptic inputs. Meanwhile, successive layers of neurons learn hierarchical causal models for the sensory input. The corresponding learning rule is local, spike-time dependent, and highly nonlinear. This approach provides a principled description of spiking and plasticity rules maximizing information transfer, while limiting

the number of costly spikes, between successive layers of neurons.

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20. Population Coding with Motion Energy Filters: The Impact of Correlations

F. Klam, Richard S. Zemel, Alexandre Pouget

https://doi.org/10.1162/neco.2008.20.1.146 | pp. 146-175

The codes obtained from the responses of large populations of neurons are known as population codes. Several studies have shown that the amount of information conveyed by such codes, and the format of this information, is highly dependent on the pattern of correlations. However, very little is known about the impact of response correlations (as found in actual cortical circuits) on neural coding. To address this problem, we investigated the properties of population codes obtained from motion energy filters, which provide one of the best models for motion selectivity in early visual areas. It is therefore likely that the correlations that arise among energy filters also arise among motion-selective neurons. We adopted an ideal observer approach to analyze filter responses to three sets of images: noisy sine gratings, random dots kinematograms, and images of natural scenes. We report that in our model, the structure of the population code varies with the type of image. We also show that for all sets of images, correlations convey a large fraction of the information: 40% to 90% of the total information. Moreover, ignoring those correlations when decoding leads to considerable information loss-from 50% to 93%, depending on the image type. Finally we show that it is important to consider a large population of motion energy filters in order to see the impact of correlations. Study of pairs of neurons, as is often done experimentally, can underestimate the effect of correlations.

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21. Feedback Decoding of Spatially Structured Population Activity in Cortical Maps

Nicholas V. Swindale

https://doi.org/10.1162/neco.2008.20.1.176 | pp. 176-204

A mechanism is proposed by which feedback pathways model spatial patterns of feedforward activity in cortical maps. The mechanism can be viewed equivalently as readout of a content-addressable memory or as decoding of a population code. The model is based on the evidence that cortical receptive fields can often be described as a separable product of functions along several dimensions, each represented in a spatially ordered map. Given this, it is shown that for an N-dimensional map, accurate modeling and decoding of x(N) feedforward activity patterns can be done with Nx fibers, N of which must be active at any one time. The proposed mechanism explains several known properties of the cortex and pyramidal neurons: (1) the integration of signals by dendrites with a narrow tangential distribution, that is, apical dendrites; (2) the presence of fast-conducting feedback projections with broad tangential distributions; (3) the multiplicative effects of attention on receptive field profiles; and (4) the existence of multiplicative interactions between subthreshold feedforward inputs to basal dendrites and inputs to apical dendrites.

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22. Chaotic Frequency Scaling in a Coupled Oscillator Model for Free Rhythmic Actions

Aaron Raftery, Joseph P. Cusumano, Dagmar Sternad

https://doi.org/10.1162/neco.2008.20.1.205 | pp. 205-226

The question of how best to model rhythmic movements at self-selected amplitude-frequency combinations, and their variability, is a long-standing issue. This study presents a systematic analysis of a coupled oscillator system that has successfully accounted for the experimental result that humans' preferred oscillation frequencies closely correspond to the linear resonance frequencies of the biomechanical limb systems, a phenomenon known as resonance tuning or frequency scaling. The dynamics of the coupled oscillator model is explored by numerical integration in different areas of its parameter space, where a period doubling route to chaotic dynamics is discovered. It is shown that even in the regions of the parameter space with chaotic solutions, the model still effectively scales to the biomechanical oscillator's natural frequency. Hence, there is a solution providing for frequency scaling in the presence of chaotic variability. The implications of these results for interpreting variability as fundamentally stochastic or chaotic are discussed.

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23. The Actor-Critic Learning Is Behind the Matching Law: Matching Versus Optimal Behaviors

Yutaka Sakai, Tomoki Fukai

https://doi.org/10.1162/neco.2008.20.1.227 | pp. 227-251

The ability to make a correct choice of behavior from various options is crucial for animals' survival. The neural basis for the choice of behavior has been attracting growing attention in research on biological and artificial neural systems. Alternative choice tasks with variable ratio (VR) and variable interval (VI) schedules of reinforcement have often been employed in studying decision making by animals and humans. In the VR schedule task, alternative choices are reinforced with different probabilities, and subjects learn to select the behavioral response rewarded more frequently. In the VI schedule task, alternative choices are reinforced at different average intervals independent of the choice frequencies, and the choice behavior follows the so-called matching law. The two policies appear robustly in subjects' choice of behavior, but the underlying neural mechanisms remain unknown. Here, we show that these seemingly different policies can appear from a common computational algorithm known as actor-critic learning. We present experimentally testable variations of the VI schedule in which the matching behavior gives only a suboptimal solution to decision making and show that the actor-critic system exhibits the matching behavior in the steady state of the learning even when the matching behavior is suboptimal. However, it is found that the matching behavior can earn approximately the same reward as the optimal one in many practical situations.

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24. Minimization of Error Functionals over Perceptron Networks

V■ra K■rková

https://doi.org/10.1162/neco.2008.20.1.252 | pp. 252-270

Supervised learning of perceptron networks is investigated as an optimization problem. It is shown that both the theoretical and the empirical error functionals achieve minima over sets of functions computable by networks with a given number n of perceptrons. Upper bounds on rates of convergence of these minima with n increasing are derived. The bounds depend on a certain regularity of training data expressed in terms of variational norms of functions interpolating the data (in the case of the empirical error) and the regression

function (in the case of the expected error). Dependence of this type of regularity on dimensionality and on magnitudes of partial derivatives is investigated. Conditions on the data, which guarantee that a good approximation of global minima of error functionals can be achieved using networks with a limited complexity, are derived. The conditions are in terms of oscillatory behavior of the data measured by the product of a function of the number of variables d, which is decreasing exponentially fast, and the maximum of the magnitudes of the squares of the L(1)-norms of the iterated partial derivatives of the order d of the regression function or some function, which interpolates the sample of the data. The results are illustrated by examples of data with small and high regularity constructed using Boolean functions and the gaussian function.

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25. An SMO Algorithm for the Potential Support Vector Machine

Tilman Knebel, Sepp Hochreiter, Klaus Obermayer

https://doi.org/10.1162/neco.2008.20.1.271 | pp. 271-287

We describe a fast sequential minimal optimization (SMO) procedure for solving the dual optimization problem of the recently proposed potential support vector machine (P-SVM). The new SMO consists of a sequence of iteration steps in which the Lagrangian is optimized with respect to either one (single SMO) or two (dual SMO) of the Lagrange multipliers while keeping the other variables fixed. An efficient selection procedure for Lagrange multipliers is given, and two heuristics for improving the SMO procedure are described: block optimization and annealing of the regularization parameter epsilon. A comparison of the variants shows that the dual SMO, including block optimization and annealing, performs efficiently in terms of computation time. In contrast to standard support vector machines (SVMs), the P-SVM is applicable to arbitrary dyadic data sets, but benchmarks are provided against libSVM's epsilon-SVR and C-SVC implementations for problems that are also solvable by standard SVM methods. For those problems, computation time of the P-SVM is comparable to or somewhat higher than the standard SVM. The number of support vectors found by the P-SVM is usually much smaller for the same generalization performance.

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26. On the Classification Capability of Sign-Constrained Perceptrons

Robert Legenstein, Wolfgang Maass

https://doi.org/10.1162/neco.2008.20.1.288 | pp. 288-309

The perceptron (also referred to as McCulloch-Pitts neuron, or linear threshold gate) is commonly used as a simplified model for the discrimination and learning capability of a biological neuron. Criteria that tell us when a perceptron can implement (or learn to implement) all possible dichotomies over a given set of input patterns are well known, but only for the idealized case, where one assumes that the sign of a synaptic weight can be switched during learning. We present in this letter an analysis of the classification capability of the biologically more realistic model of a sign-constrained perceptron, where the signs of synaptic weights remain fixed during learning (which is the case for most types of biological synapses). In particular, the VC-dimension of sign-constrained perceptrons is determined, and a necessary and sufficient criterion is provided that tells us when all 2(m) dichotomies over a given set of m patterns can be learned by a sign-constrained perceptron. We also show that uniformity of L(1) norms of input patterns is a sufficient condition for full representation power in the case where all weights are required to be nonnegative. Finally, we exhibit cases where the sign constraint of a perceptron drastically reduces its classification capability.

Our theoretical analysis is complemented by computer simulations, which demonstrate in particular that sparse input patterns improve the classification capability of sign-constrained perceptrons.

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27. The Astrocyte as a Gatekeeper of Synaptic Information Transfer

Vladislav Volman, Eshel Ben■Jacob, Herbert Levine

https://doi.org/10.1162/neco.2007.19.2.303 | pp. 303-326

We present a simple biophysical model for the coupling between synaptic transmission and the local calcium concentration on an enveloping astrocytic domain. This interaction enables the astrocyte to modulate the information flow from presynaptic to postsynaptic cells in a manner dependent on previous activity at this and other nearby synapses. Our model suggests a novel, testable hypothesis for the spike timing statistics measured for rapidly firing cells in culture experiments.

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28. Inhomogeneities in Heteroassociative Memories with Linear Learning Rules

David C. Sterratt, David Willshaw

https://doi.org/10.1162/neco.2007.08-06-301 | pp. 311-344

We investigate how various inhomogeneities present in synapses and neurons affect the performance of feedforward associative memories with linear learning, a high-level network model of hippocampal circuitry and plasticity. The inhomogeneities incorporated into the model are differential input attenuation, stochastic synaptic transmission, and memories learned with varying intensity. For a class of local learning rules, we determine the memory capacity of the model by extending previous analysis. We find that the signal-to-noise ratio (SNR), a measure of fidelity of recall, depends on the coefficients of variation (CVs) of the attenuation factors, the transmission variables, and the intensity of the memories, as well as the parameters of the learning rule, pattern sparsity and the number of memories stored. To predict the effects of attenuation due to extended dendritic trees, we use distributions of attenuations appropriate to unbranched and branched dendritic trees. Biological parameters for stochastic transmission are used to determine the CV of the transmission factors. The reduction in SNR due to differential attenuation is surprisingly low compared to the reduction due to stochastic transmission. Training a network by storing memories at different intensities is equivalent to using a learning rule incorporating weight decay. In this type of network, new memories can be stored continuously at the expense of older ones being forgotten (a palimpsest). We show that there is an optimal rate of weight decay that maximizes the capacity of the network, which is a factor of e lower than its nonpalimpsest equivalent.

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29. Thermodynamically Equivalent Silicon Models of Voltage-Dependent Ion Channels

K.M. Hynna, Kwabena Boahen

https://doi.org/10.1162/neco.2007.19.2.327 | pp. 327-350

We model ion channels in silicon by exploiting similarities between the thermodynamic principles that govern ion channels and those that govern transistors. Using just eight transistors, we replicate--for the first time in

silicon--the sigmoidal voltage dependence of activation (or inactivation) and the bell-shaped voltage-dependence of its time constant. We derive equations describing the dynamics of our silicon analog and explore its flexibility by varying various parameters. In addition, we validate the design by implementing a channel with a single activation variable. The design's compactness allows tens of thousands of copies to be built on a single chip, facilitating the study of biologically realistic models of neural computation at the network level in silicon.

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30. A Neural Network Model of the Eriksen Task: Reduction, Analysis, and Data Fitting

Yuan Sophie Liu, Philip Holmes, Jonathan D. Cohen

https://doi.org/10.1162/neco.2007.08-06-313 | pp. 345-373

We analyze a neural network model of the Eriksen task: a two-alternative forced-choice task in which subjects must correctly identify a central stimulus and disregard flankers that may or may not be compatible with it. We linearize and decouple the model, deriving a reduced drift-diffusion process with variable drift rate that describes the accumulation of net evidence in favor of either alternative, and we use this to analytically describe how accuracy and response time data depend on model parameters. Such analyses both assist parameter tuning in network models and suggest explanations of changing drift rates in terms of attention. We compare our results with numerical simulations of the full nonlinear model and with empirical data and show good fits to both with fewer parameters.

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31. Spatiotemporal Conversion of Auditory Information for Cochleotopic Mapping

Osamu Hoshino

https://doi.org/10.1162/neco.2007.19.2.351 | pp. 351-370

Auditory communication signals such as monkey calls are complex FM vocal sounds and in general induce action potentials in different timing in the primary auditory cortex. Delay line scheme is one of the effective ways for detecting such neuronal timing. However, the scheme is not straightforwardly applicable if the time intervals of signals are beyond the latency time of delay lines. In fact, monkey calls are often expressed in longer time intervals (hundreds of milliseconds to seconds) and are beyond the latency times observed in the brain (less than several hundreds of milliseconds). Here, we propose a cochleotopic map similar to that in vision known as a retinotopic map. We show that information about monkey calls could be mapped on a cochleotopic cortical network as spatiotemporal firing patterns of neurons, which can then be decomposed into simple (linearly sweeping) FM components and integrated into unified percepts by higher cortical networks. We suggest that the spatiotemporal conversion of auditory information may be essential for developing the cochleotopic map, which could serve as the foundation for later processing, or monkey call identification by higher cortical areas.

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32. Reducing the Variability of Neural Responses: A Computational Theory of Spike-Timing-Dependent Plasticity

Sander M. Bohté, Michael C. Mozer

https://doi.org/10.1162/neco.2007.19.2.371 | pp. 371-403

Experimental studies have observed synaptic potentiation when a presynaptic neuron fires shortly before a postsynaptic neuron and synaptic depression when the presynaptic neuron fires shortly after. The dependence of synaptic modulation on the precise timing of the two action potentials is known as spike-timing dependent plasticity (STDP). We derive STDP from a simple computational principle: synapses adapt so as to minimize the postsynaptic neuron's response variability to a given presynaptic input, causing the neuron's output to become more reliable in the face of noise. Using an objective function that minimizes response variability and the biophysically realistic spike-response model of Gerstner (2001), we simulate neurophysiological experiments and obtain the characteristic STDP curve along with other phenomena, including the reduction in synaptic plasticity as synaptic efficacy increases. We compare our account to other efforts to derive STDP from computational principles and argue that our account provides the most comprehensive coverage of the phenomena. Thus, reliability of neural response in the face of noise may be a key goal of unsupervised cortical adaptation.

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33. Second-Order SMO Improves SVM Online and Active Learning

Tobias Glasmachers, Christian Igel

https://doi.org/10.1162/neco.2007.10-06-354 | pp. 374-382

Iterative learning algorithms that approximate the solution of support vector machines (SVMs) have two potential advantages. First, they allow online and active learning. Second, for large data sets, computing the exact SVM solution may be too time-consuming, and an efficient approximation can be preferable. The powerful LASVM iteratively approaches the exact SVM solution using sequential minimal optimization (SMO). It allows efficient online and active learning. Here, this algorithm is considerably improved in speed and accuracy by replacing the working set selection in the SMO steps. A second-order working set selection strategy, which greedily aims at maximizing the progress in each single step, is incorporated.

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34. Gamma Oscillations and Stimulus Selection

Christoph Börgers, Nancy Kopell

https://doi.org/10.1162/neco.2007.07-06-289 | pp. 383-414

More coherent excitatory stimuli are known to have a competitive advantage over less coherent ones. We show here that this advantage is amplified greatly when the target includes inhibitory interneurons acting via GABA(A)-receptor-mediated synapses and the coherent input oscillates at gamma frequency. We hypothesize that therein lies, at least in part, the functional significance of the experimentally observed link between attentional biasing of stimulus competition and gamma frequency rhythmicity.

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35. Fast Population Coding

Quentin J. M. Huys, Richard S. Zemel, Rama Natarajan, Peter Dayan

https://doi.org/10.1162/neco.2007.19.2.404 | pp. 404-441

Uncertainty coming from the noise in its neurons and the ill-posed nature of many tasks plagues neural computations. Maybe surprisingly, many studies show that the brain manipulates these forms of uncertainty in a probabilistically consistent and normative manner, and there is now a rich theoretical literature on the capabilities of populations of neurons to implement computations in the face of uncertainty. However, one major facet of uncertainty has received comparatively little attention: time. In a dynamic, rapidly changing world, data are only temporarily relevant. Here, we analyze the computational consequences of encoding stimulus trajectories in populations of neurons. For the most obvious, simple, instantaneous encoder, the correlations induced by natural, smooth stimuli engender a decoder that requires access to information that is nonlocal both in time and across neurons. This formally amounts to a ruinous representation. We show that there is an alternative encoder that is computationally and representationally powerful in which each spike contributes independent information; it is independently decodable, in other words. We suggest this as an appropriate foundation for understanding time-varying population codes. Furthermore, we show how adaptation to temporal stimulus statistics emerges directly from the demands of simple decoding.

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36. STDP Provides the Substrate for Igniting Synfire Chains by Spatiotemporal Input Patterns

Ryosuke Hosaka, Osamu Araki, Tohru Ikeguchi

https://doi.org/10.1162/neco.2007.11-05-043 | pp. 415-435

Spike-timing-dependent synaptic plasticity (STDP), which depends on the temporal difference between pre- and postsynaptic action potentials, is observed in the cortices and hippocampus. Although several theoretical and experimental studies have revealed its fundamental aspects, its functional role remains unclear. To examine how an input spatiotemporal spike pattern is altered by STDP, we observed the output spike patterns of a spiking neural network model with an asymmetrical STDP rule when the input spatiotemporal pattern is repeatedly applied. The spiking neural network comprises excitatory and inhibitory neurons that exhibit local interactions. Numerical experiments show that the spiking neural network generates a single global synchrony whose relative timing depends on the input spatiotemporal pattern and the neural network structure. This result implies that the spiking neural network learns the transformation from spatiotemporal to temporal information. In the literature, the origin of the synfire chain has not been sufficiently focused on. Our results indicate that spiking neural networks with STDP can ignite synfire chains in the cortices.

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37. A-Type K⁺ Current Can Act as a Trigger for Bursting in the Absence of a Slow Variable

Natalia Toporikova, Joël Tabak, Marc E. Freeman, Richard Bertram

https://doi.org/10.1162/neco.2007.08-06-310 | pp. 436-451

Models of bursting in single cells typically include two subsystems with different timescales. Variations in one or more slow variables switch the system between a silent and a spiking state. We have developed a model for bursting in the pituitary lactotroph that does not include any slow variable. The model incorporates fast, noninactivating calcium and potassium currents (the spike-generating mechanism), as well as the fast, inactivating A-type potassium current (I(A)). I(A) is active only briefly at the beginning of a burst, but this brief impulse of I(A) acts as a burst trigger, injecting the spike trajectory close to an unstable steady state. The

spiraling of the trajectory away from the steady state produces a period of low-amplitude spiking typical of lactotrophs. Increasing the conductance of A-type potassium current brings the trajectory closer to the unstable steady state, increasing burst duration. However, this also increases interburst interval, and for larger conductance values, all activity stops. To our knowledge, this is the first example of a physiologically based, single-compartmental model of bursting with no slow subsystem.

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38. The Basal Ganglia and Cortex Implement Optimal Decision Making Between Alternative Actions

Rafa■ Bogacz, Kevin Gurney

https://doi.org/10.1162/neco.2007.19.2.442 | pp. 442-477

Neurophysiological studies have identified a number of brain regions critically involved in solving the problem of action selection or decision making. In the case of highly practiced tasks, these regions include cortical areas hypothesized to integrate evidence supporting alternative actions and the basal ganglia, hypothesized to act as a central switch in gating behavioral requests. However, despite our relatively detailed knowledge of basal ganglia biology and its connectivity with the cortex and numerical simulation studies demonstrating selective function, no formal theoretical framework exists that supplies an algorithmic description of these circuits. This article shows how many aspects of the anatomy and physiology of the circuit involving the cortex and basal ganglia are exactly those required to implement the computation defined by an asymptotically optimal statistical test for decision making: the multihypothesis sequential probability ratio test (MSPRT). The resulting model of basal ganglia provides a new framework for understanding the computation in the basal ganglia during decision making in highly practiced tasks. The predictions of the theory concerning the properties of particular neuronal populations are validated in existing experimental data. Further, we show that this neurobiologically grounded implementation of MSPRT outperforms other candidates for neural decision making, that it is structurally and parametrically robust, and that it can accommodate cortical mechanisms for decision making in a way that complements those in basal ganglia.

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39. Design of Continuous Attractor Networks with Monotonic Tuning Using a Symmetry Principle

Christian K. Machens, Carlos D. Brody

https://doi.org/10.1162/neco.2007.07-06-297 | pp. 452-485

Neurons that sustain elevated firing in the absence of stimuli have been found in many neural systems. In graded persistent activity, neurons can sustain firing at many levels, suggesting a widely found type of network dynamics in which networks can relax to any one of a continuum of stationary states. The reproduction of these findings in model networks of nonlinear neurons has turned out to be nontrivial. A particularly insightful model has been the "bump attractor," in which a continuous attractor emerges through an underlying symmetry in the network connectivity matrix. This model, however, cannot account for data in which the persistent firing of neurons is a monotonic -- rather than a bell-shaped -- function of a stored variable. Here, we show that the symmetry used in the bump attractor network can be employed to create a whole family of continuous attractor networks, including those with monotonic tuning. Our design is based on tuning the external inputs to networks that have a connectivity matrix with Toeplitz symmetry. In particular,

we provide a complete analytical solution of a line attractor network with monotonic tuning and show that for many other networks, the numerical tuning of synaptic weights reduces to the computation of a single parameter.

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40. Realistically Coupled Neural Mass Models Can Generate EEG Rhythms

Roberto C. Sotero, Nelson J. Trujillo■Barreto, Yasser Iturria■Medina, Félix Carbonell, Juan Carlos Jiménez
https://doi.org/10.1162/neco.2007.19.2.478 | pp. 478-512

We study the generation of EEG rhythms by means of realistically coupled neural mass models. Previous neural mass models were used to model cortical voxels and the thalamus. Interactions between voxels of the same and other cortical areas and with the thalamus were taken into account. Voxels within the same cortical area were coupled (short-range connections) with both excitatory and inhibitory connections, while coupling between areas (long-range connections) was considered to be excitatory only. Short-range connection strengths were modeled by using a connectivity function depending on the distance between voxels. Coupling strength parameters between areas were defined from empirical anatomical data employing the information obtained from probabilistic paths, which were tracked by water diffusion imaging techniques and used to quantify white matter tracts in the brain. Each cortical voxel was then described by a set of 16 random differential equations, while the thalamus was described by a set of 12 random differential equations. Thus, for analyzing the neuronal dynamics emerging from the interaction of several areas, a large system of differential equations needs to be solved. The sparseness of the estimated anatomical connectivity matrix reduces the number of connection parameters substantially, making the solution of this system faster. Simulations of human brain rhythms were carried out in order to test the model. Physiologically plausible results were obtained based on this anatomically constrained neural mass model.

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41. Brain Reading Using Full Brain Support Vector Machines for Object Recognition: There Is No “Face” Identification Area

Stephen José Hanson, Yaroslav O. Halchenko
https://doi.org/10.1162/neco.2007.09-06-340 | pp. 486-503

Over the past decade, object recognition work has confounded voxel response detection with potential voxel class identification. Consequently, the claim that there are areas of the brain that are necessary and sufficient for object identification cannot be resolved with existing associative methods (e.g., the general linear model) that are dominant in brain imaging methods. In order to explore this controversy we trained full brain (40,000 voxels) single TR (repetition time) classifiers on data from 10 subjects in two different recognition tasks on the most controversial classes of stimuli (house and face) and show 97.4% median out-of-sample (unseen TRs) generalization. This performance allowed us to reliably and uniquely assay the classifier's voxel diagnosticity in all individual subjects' brains. In this two-class case, there may be specific areas diagnostic for house stimuli (e.g., LO) or for face stimuli (e.g., STS); however, in contrast to the detection results common in this literature, neither the fusiform face area nor parahippocampal place area is shown to be uniquely diagnostic for faces or places, respectively.

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42. Neural Network Models for Conditional Distribution Under Bayesian Analysis

Tatiana Miazhynskaia, Sylvia Frühwirth■Schnatter, Georg Dorffner

https://doi.org/10.1162/neco.2007.3182 | pp. 504-522

We use neural networks (NN) as a tool for a nonlinear autoregression to predict the second moment of the conditional density of return series. The NN models are compared to the popular econometric GARCH(1,1) model. We estimate the models in a Bayesian framework using Markov chain Monte Carlo posterior simulations. The interlinked aspects of the proposed Bayesian methodology are identification of NN hidden units and treatment of NN complexity based on model evidence. The empirical study includes the application of the designed strategy to market data, where we found a strong support for a nonlinear multilayer perceptron model with two hidden units.

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43. Dimension Selection for Feature Selection and Dimension Reduction with Principal and Independent Component Analysis

Inge Koch, Kanta Naito

https://doi.org/10.1162/neco.2007.19.2.513 | pp. 513-545

This letter is concerned with the problem of selecting the best or most informative dimension for dimension reduction and feature extraction in high-dimensional data. The dimension of the data is reduced by principal component analysis; subsequent application of independent component analysis to the principal component scores determines the most nongaussian directions in the lower-dimensional space. A criterion for choosing the optimal dimension based on bias-adjusted skewness and kurtosis is proposed. This new dimension selector is applied to real data sets and compared to existing methods. Simulation studies for a range of densities show that the proposed method performs well and is more appropriate for nongaussian data than existing methods.

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44. Efficient Tracking of the Dominant Eigenspace of a Normalized Kernel Matrix

Geert Gins, Ilse Smets, Jan Van Impe

https://doi.org/10.1162/neco.2007.05-06-213 | pp. 523-554

Various machine learning problems rely on kernel-based methods. The power of these methods resides in the ability to solve highly nonlinear problems by reformulating them in a linear context. The dominant eigenspace of a (normalized) kernel matrix is often required. Unfortunately, the computational requirements of the existing kernel methods are such that the applicability is restricted to relatively small data sets. This letter therefore focuses on a kernel-based method for large data sets. More specifically, a numerically stable tracking algorithm for the dominant eigenspace of a normalized kernel matrix is proposed, which proceeds by an updating (the addition of a new data point) followed by a downdating (the exclusion of an old data point) of the kernel matrix. Testing the algorithm on some representative case studies reveals that a very good approximation of the dominant eigenspace is obtained, while only a minimal amount of operations and memory space per iteration step is required.

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45. One-Bit-Matching Theorem for ICA, Convex-Concave Programming on Polyhedral Set, and Distribution Approximation for Combinatorics

Lei Xu

https://doi.org/10.1162/neco.2007.19.2.546 | pp. 546-569

According to the proof by Liu, Chiu, and Xu (2004) on the so-called one-bit-matching conjecture (Xu, Cheung, and Amari, 1998a), all the sources can be separated as long as there is an one-to-one same-sign correspondence between the kurtosis signs of all source probability density functions (pdf's) and the kurtosis signs of all model pdf's, which is widely believed and implicitly supported by many empirical studies. However, this proof is made only in a weak sense that the conjecture is true when the global optimal solution of an independent component analysis criterion is reached. Thus, it cannot support the successes of many existing iterative algorithms that usually converge at one of the local optimal solutions. This article presents a new mathematical proof that is obtained in a strong sense that the conjecture is also true when any one of local optimal solutions is reached in helping to investigating convex-concave programming on a polyhedral set. Theorems are also provided not only on partial separation of sources when there is a partial matching between the kurtosis signs, but also on an interesting duality of maximization and minimization on source separation. Moreover, corollaries are obtained on an interesting duality, with supergaussian sources separated by maximization and subgaussian sources separated by minimization. Also, a corollary is obtained to confirm the symmetric orthogonalization implementation of the kurtosis extreme approach for separating multiple sources in parallel, which works empirically but lacks mathematical proof. Furthermore, a linkage has been set up to combinatorial optimization from a distribution approximation perspective and a Stiefel manifold perspective, with algorithms that guarantee convergence as well as satisfaction of constraints.

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46. Robust L1 Principal Component Analysis and Its Bayesian Variational Inference

Junbin Gao

https://doi.org/10.1162/neco.2007.11-06-397 | pp. 555-572

We introduce a robust probabilistic L1-PCA model in which the conventional gaussian distribution for the noise in the observed data was replaced by the Laplacian distribution (or L1 distribution). Due to the heavy tail characteristics of the L1 distribution, the proposed model is supposed to be more robust against data outliers. In this letter, we demonstrate how a variational approximation scheme enables effective inference of key parameters in the probabilistic L1-PCA model. As the L1 density can be expanded as a superposition of infinite number of gaussian densities, we express the L1-PCA model as a marginalized model over the superpositions. By doing so, a tractable Bayesian inference can be achieved based on the variational expectation-maximization-type algorithm.

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47. Boundedness and Stability for Integrodifferential Equations Modeling Neural Field with Time Delay

Xuyang Lou, Baotong Cui

https://doi.org/10.1162/neco.2007.19.2.570 | pp. 570-581

In this letter, delayed integrodifferential equations modeling neural field (DIEMNF) is studied. The model of DIEMNF is first established as a modified neural field model. Second, it has been proved that if the interconnection of neural field is symmetric, then every trajectory of the system converges to an equilibrium. The boundedness condition for integrodifferential equations modeling neural field with delay is obtained. Moreover, when the interconnection is asymmetric, we give a sufficient condition that can guarantee that the equilibrium of the DIEMNF is unique and is also a global attractor.

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48. Generation of Optimal Artificial Neural Networks Using a Pattern Search Algorithm: Application to Approximation of Chemical Systems

Matthias Ihme, Alison L. Marsden, Heinz Pitsch

https://doi.org/10.1162/neco.2007.08-06-316 | pp. 573-601

A pattern search optimization method is applied to the generation of optimal artificial neural networks (ANNs). Optimization is performed using a mixed variable extension to the generalized pattern search method. This method offers the advantage that categorical variables, such as neural transfer functions and nodal connectivities, can be used as parameters in optimization. When used together with a surrogate, the resulting algorithm is highly efficient for expensive objective functions. Results demonstrate the effectiveness of this method in optimizing an ANN for the number of neurons, the type of transfer function, and the connectivity among neurons. The optimization method is applied to a chemistry approximation of practical relevance. In this application, temperature and a chemical source term are approximated as functions of two independent parameters using optimal ANNs. Comparison of the performance of optimal ANNs with conventional tabulation methods demonstrates equivalent accuracy by considerable savings in memory storage. The architecture of the optimal ANN for the approximation of the chemical source term consists of a fully connected feedforward network having four nonlinear hidden layers and 117 synaptic weights. An equivalent representation of the chemical source term using tabulation techniques would require a 500 x 500 grid point discretization of the parameter space.

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49. Temporal Symmetry in Primary Auditory Cortex: Implications for Cortical Connectivity

Jonathan Z. Simon, Didier A. Depireux, David J. Klein, Jonathan B. Fritz, Shihab Shamma

https://doi.org/10.1162/neco.2007.19.3.583 | pp. 583-638

Neurons in primary auditory cortex (AI) in the ferret (Mustela putorius) that are well described by their spectrotemporal response field (STRF) are found also to have a distinctive property that we call temporal symmetry. For temporally symmetric neurons, every temporal cross-section of the STRF (impulse response) is given by the same function of time, except for a scaling and a Hilbert rotation. This property held in 85% of neurons (123 out of 145) recorded from awake animals and in 96% of neurons (70 out of 73) recorded from anesthetized animals. This property of temporal symmetry is highly constraining for possible models of functional neural connectivity within and into AI. We find that the simplest models of functional thalamic input, from the ventral medial geniculate body (MGB), into the entry layers of AI are ruled out because they are incompatible with the constraints of the observed temporal symmetry. This is also the case for the simplest models of functional intracortical connectivity. Plausible models that do generate temporal symmetry, from both thalamic and intracortical inputs, are presented. In particular, we propose that two

specific characteristics of the thalamocortical interface may be responsible. The first is a temporal mismatch between the fast dynamics of the thalamus and the slow responses of the cortex. The second is that all thalamic inputs into a cortical module (or a cluster of cells) must be restricted to one point of entry (or one cell in the cluster). This latter property implies a lack of correlated horizontal interactions across cortical modules during the STRF measurements. The implications of these insights in the auditory system, and comparisons with similar properties in the visual system, are explored.

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50. A Sensorimotor Approach to Sound Localization

Murat Aytekin, Cynthia F. Moss, Jonathan Z. Simon
https://doi.org/10.1162/neco.2007.12-05-094 | pp. 603-635

Sound localization is known to be a complex phenomenon, combining multisensory information processing, experience-dependent plasticity, and movement. Here we present a sensorimotor model that addresses the question of how an organism could learn to localize sound sources without any a priori neural representation of its head-related transfer function or prior experience with auditory spatial information. We demonstrate quantitatively that the experience of the sensory consequences of its voluntary motor actions allows an organism to learn the spatial location of any sound source. Using examples from humans and echolocating bats, our model shows that a naive organism can learn the auditory space based solely on acoustic inputs and their relation to motor states.

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51. A Note on Lewicki-Sejnowski Gradient for Learning Overcomplete Representations

Zhaoshui He, Shengli Xie, Liqing Zhang, Andrzej Cichocki
https://doi.org/10.1162/neco.2007.07-06-296 | pp. 636-643

Overcomplete representations have greater robustness in noise environment and also have greater flexibility in matching structure in the data. Lewicki and Sejnowski (2000) proposed an efficient extended natural gradient for learning the overcomplete basis and developed an overcomplete representation approach. However, they derived their gradient by many approximations, and their proof is very complicated. To give a stronger theoretical basis, we provide a brief and more rigorous mathematical proof for this gradient in this note. In addition, we propose a more robust constrained Lewicki-Sejnowski gradient.

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52. Optimality Model of Unsupervised Spike-Timing-Dependent Plasticity: Synaptic Memory and Weight Distribution

Taro Toyoizumi, Jean-Pascal Pfister, Kazuyuki Aihara, Wulfram Gerstner
https://doi.org/10.1162/neco.2007.19.3.639 | pp. 639-671

We studied the hypothesis that synaptic dynamics is controlled by three basic principles: (1) synapses adapt their weights so that neurons can effectively transmit information, (2) homeostatic processes stabilize the mean firing rate of the postsynaptic neuron, and (3) weak synapses adapt more slowly than strong ones, while maintenance of strong synapses is costly. Our results show that a synaptic update rule derived from these principles shares features, with spike-timing-dependent plasticity, is sensitive to correlations in the

input and is useful for synaptic memory. Moreover, input selectivity (sharply tuned receptive fields) of postsynaptic neurons develops only if stimuli with strong features are presented. Sharply tuned neurons can coexist with unselective ones, and the distribution of synaptic weights can be unimodal or bimodal. The formulation of synaptic dynamics through an optimality criterion provides a simple graphical argument for the stability of synapses, necessary for synaptic memory.

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53. Valuations for Spike Train Prediction

Vladimir Itskov, Carina Curto, Kenneth D. Harris

https://doi.org/10.1162/neco.2007.3179 | pp. 644-667

The ultimate product of an electrophysiology experiment is often a decision on which biological hypothesis or model best explains the observed data. We outline a paradigm designed for comparison of different models, which we refer to as spike train prediction. A key ingredient of this paradigm is a prediction quality valuation that estimates how close a predicted conditional intensity function is to an actual observed spike train. Although a valuation based on log likelihood (L) is most natural, it has various complications in this context. We propose that a quadratic valuation (Q) can be used as an alternative to L. Q shares some important theoretical properties with L, including consistency, and the two valuations perform similarly on simulated and experimental data. Moreover, Q is more robust than L, and optimization with Q can dramatically improve computational efficiency. We illustrate the utility of Q for comparing models of peer prediction, where it can be computed directly from cross-correlograms. Although Q does not have a straightforward probabilistic interpretation, Q is essentially given by Euclidean distance.

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54. How Optimal Stimuli for Sensory Neurons Are Constrained by Network Architecture

Christopher DiMattina, Kechen Zhang

https://doi.org/10.1162/neco.2007.11-05-076 | pp. 668-708

Identifying the optimal stimuli for a sensory neuron is often a difficult process involving trial and error. By analyzing the relationship between stimuli and responses in feedforward and stable recurrent neural network models, we find that the stimulus yielding the maximum firing rate response always lies on the topological boundary of the collection of all allowable stimuli, provided that individual neurons have increasing input-output relations or gain functions and that the synaptic connections are convergent between layers with nondegenerate weight matrices. This result suggests that in neurophysiological experiments under these conditions, only stimuli on the boundary need to be tested in order to maximize the response, thereby potentially reducing the number of trials needed for finding the most effective stimuli. Even when the gain functions allow firing rate cutoff or saturation, a peak still cannot exist in the stimulus-response relation in the sense that moving away from the optimum stimulus always reduces the response. We further demonstrate that the condition for nondegenerate synaptic connections also implies that proper stimuli can independently perturb the activities of all neurons in the same layer. One example of this type of manipulation is changing the activity of a single neuron in a given processing layer while keeping that of all others constant. Such stimulus perturbations might help experimentally isolate the interactions of selected neurons within a network.

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55. Nonparametric Modeling of Neural Point Processes via Stochastic Gradient Boosting Regression

Wilson Truccolo, John P. Donoghue

https://doi.org/10.1162/neco.2007.19.3.672 | pp. 672-705

Statistical nonparametric modeling tools that enable the discovery and approximation of functional forms (e.g., tuning functions) relating neural spiking activity to relevant covariates are desirable tools in neuroscience. In this article, we show how stochastic gradient boosting regression can be successfully extended to the modeling of spiking activity data while preserving their point process nature, thus providing a robust nonparametric modeling tool. We formulate stochastic gradient boosting in terms of approximating the conditional intensity function of a point process in discrete time and use the standard likelihood of the process to derive the loss function for the approximation problem. To illustrate the approach, we apply the algorithm to the modeling of primary motor and parietal spiking activity as a function of spiking history and kinematics during a two-dimensional reaching task. Model selection, goodness of fit via the time rescaling theorem, model interpretation via partial dependence plots, ranking of covariates according to their relative importance, and prediction of peri-event time histograms are illustrated and discussed. Additionally, we use the tenfold cross-validated log likelihood of the modeled neural processes (67 cells) to compare the performance of gradient boosting regression to two alternative approaches: standard generalized linear models (GLMs) and Bayesian P-splines with Markov chain Monte Carlo (MCMC) sampling. In our data set, gradient boosting outperformed both Bayesian P-splines (in approximately 90% of the cells) and GLMs (100%). Because of its good performance and computational efficiency, we propose stochastic gradient boosting regression as an off-the-shelf nonparametric tool for initial analyses of large neural data sets (e.g., more than 50 cells; more than 10(5) samples per cell) with corresponding multidimensional covariate spaces (e.g., more than four covariates). In the cases where a functional form might be amenable to a more compact representation, gradient boosting might also lead to the discovery of simpler, parametric models.

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56. Synchrony of Neuronal Oscillations Controlled by GABAergic Reversal Potentials

Ho Young Jeong, Boris Gutkin

https://doi.org/10.1162/neco.2007.19.3.706 | pp. 706-729

GABAergic synapse reversal potential is controlled by the concentration of chloride. This concentration can change significantly during development and as a function of neuronal activity. Thus, GABA inhibition can be hyperpolarizing, shunting, or partially depolarizing. Previous results pinpointed the conditions under which hyperpolarizing inhibition (or depolarizing excitation) can lead to synchrony of neural oscillators. Here we examine the role of the GABAergic reversal potential in generation of synchronous oscillations in circuits of neural oscillators. Using weakly coupled oscillator analysis, we show when shunting and partially depolarizing inhibition can produce synchrony, asynchrony, and coexistence of the two. In particular, we show that this depends critically on such factors as the firing rate, the speed of the synapse, spike frequency adaptation, and, most important, the dynamics of spike generation (type I versus type II). We back up our analysis with simulations of small circuits of conductance-based neurons, as well as large-scale networks of neural oscillators. The simulation results are compatible with the analysis: for example, when bistability is predicted analytically, the large-scale network shows clustered states.

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57. A Principle for Learning Egocentric-Allocentric Transformation

Patrick Byrne, Suzanna Becker

https://doi.org/10.1162/neco.2007.10-06-361 | pp. 709-737

Numerous single-unit recording studies have found mammalian hippocampal neurons that fire selectively for the animal's location in space, independent of its orientation. The population of such neurons, commonly known as place cells, is thought to maintain an allocentric, or orientation-independent, internal representation of the animal's location in space, as well as mediating long-term storage of spatial memories. The fact that spatial information from the environment must reach the brain via sensory receptors in an inherently egocentric, or viewpoint-dependent, fashion leads to the question of how the brain learns to transform egocentric sensory representations into allocentric ones for long-term memory storage. Additionally, if these long-term memory representations of space are to be useful in guiding motor behavior, then the reverse transformation, from allocentric to egocentric coordinates, must also be learned. We propose that orientation-invariant representations can be learned by neural circuits that follow two learning principles: minimization of reconstruction error and maximization of representational temporal inertia. Two different neural network models are presented that adhere to these learning principles, the first by direct optimization through gradient descent and the second using a more biologically realistic circuit based on the restricted Boltzmann machine (Hinton, 2002; Smolensky, 1986). Both models lead to orientation-invariant representations, with the latter demonstrating place-cell-like responses when trained on a linear track environment.

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58. Reinforcement Learning State Estimator

Jun Morimoto, Kenji Doya

https://doi.org/10.1162/neco.2007.19.3.730 | pp. 730-756

In this study, we propose a novel use of reinforcement learning for estimating hidden variables and parameters of nonlinear dynamical systems. A critical issue in hidden-state estimation is that we cannot directly observe estimation errors. However, by defining errors of observable variables as a delayed penalty, we can apply a reinforcement learning frame-work to state estimation problems. Specifically, we derive a method to construct a nonlinear state estimator by finding an appropriate feedback input gain using the policy gradient method. We tested the proposed method on single pendulum dynamics and show that the joint angle variable could be successfully estimated by observing only the angular velocity, and vice versa. In addition, we show that we could acquire a state estimator for the pendulum swing-up task in which a swing-up controller is also acquired by reinforcement learning simultaneously. Furthermore, we demonstrate that it is possible to estimate the dynamics of the pendulum itself while the hidden variables are estimated in the pendulum swing-up task. Application of the proposed method to a two-linked biped model is also presented.

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59. Bayesian Model Comparison in Nonlinear BOLD fMRI Hemodynamics

Daniel J. Jacobsen, Lars Kai Hansen, Kristoffer H. Madsen

https://doi.org/10.1162/neco.2007.07-06-282 | pp. 738-755

Nonlinear hemodynamic models express the BOLD (blood oxygenation level dependent) signal as a nonlinear, parametric functional of the temporal sequence of local neural activity. Several models have been proposed for both the neural activity and the hemodynamics. We compare two such combined models: the original balloon model with a square-pulse neural model (Friston, Mechelli, Turner, & Price, 2000) and an extended balloon model with a more sophisticated neural model (Buxton, Uludag, Dubowitz, & Liu, 2004). We learn the parameters of both models using a Bayesian approach, where the distribution of the parameters conditioned on the data is estimated using Markov chain Monte Carlo techniques. Using a split-half resampling procedure (Strother, Anderson, & Hansen, 2002), we compare the generalization abilities of the models as well as their reproducibility, for both synthetic and real data, recorded from two different visual stimulation paradigms. The results show that the simple model is the better one for these data.

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60. Implications of Noise and Neural Heterogeneity for Vestibulo-Ocular Reflex Fidelity

Timothy M. Hospedales, Mark C. W. van Rossum, Bruce Graham, Mayank B. Dutia
https://doi.org/10.1162/neco.2007.09-06-339 | pp. 756-778

The vestibulo-ocular reflex (VOR) is characterized by a short-latency, high-fidelity eye movement response to head rotations at frequencies up to 20 Hz. Electrophysiological studies of medial vestibular nucleus (MVN) neurons, however, show that their response to sinusoidal currents above 10 to 12 Hz is highly nonlinear and distorted by aliasing for all but very small current amplitudes. How can this system function in vivo when single cell response cannot explain its operation? Here we show that the necessary wide VOR frequency response may be achieved not by firing rate encoding of head velocity in single neurons, but in the integrated population response of asynchronously firing, intrinsically active neurons. Diffusive synaptic noise and the pacemaker-driven, intrinsic firing of MVN cells synergistically maintain asynchronous, spontaneous spiking in a population of model MVN neurons over a wide range of input signal amplitudes and frequencies. Response fidelity is further improved by a reciprocal inhibitory link between two MVN populations, mimicking the vestibular commissural system in vivo, but only if asynchrony is maintained by noise and pacemaker inputs. These results provide a previously missing explanation for the full range of VOR function and a novel account of the role of the intrinsic pacemaker conductances in MVN cells. The values of diffusive noise and pacemaker currents that give optimal response fidelity yield firing statistics similar to those in vivo, suggesting that the in vivo network is tuned to optimal performance. While theoretical studies have argued that noise and population heterogeneity can improve coding, to our knowledge this is the first evidence indicating that these parameters are indeed tuned to optimize coding fidelity in a neural control system in vivo.

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61. Training Recurrent Networks by Evolino

Jürgen Schmidhuber, Daan Wierstra, Matteo Gagliolo, Faustino Gomez
https://doi.org/10.1162/neco.2007.19.3.757 | pp. 757-779

In recent years, gradient-based LSTM recurrent neural networks (RNNs) solved many previously RNN-unlearnable tasks. Sometimes, however, gradient information is of little use for training RNNs, due to numerous local minima. For such cases, we present a novel method: EVOlution of systems with LINear

Outputs (Evolino). Evolino evolves weights to the nonlinear, hidden nodes of RNNs while computing optimal linear mappings from hidden state to output, using methods such as pseudo-inverse-based linear regression. If we instead use quadratic programming to maximize the margin, we obtain the first evolutionary recurrent support vector machines. We show that Evolino-based LSTM can solve tasks that Echo State nets (Jaeger, 2004a) cannot and achieves higher accuracy in certain continuous function generation tasks than conventional gradient descent RNNs, including gradient-based LSTM.

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62. Minimum Acceleration Criterion with Constraints Implies Bang-Bang Control as an Underlying Principle for Optimal Trajectories of Arm Reaching Movements

Shay Ben-Itzhak, Amir Karniel

https://doi.org/10.1162/neco.2007.12-05-077 | pp. 779-812

Rapid arm-reaching movements serve as an excellent test bed for any theory about trajectory formation. How are these movements planned? A minimum acceleration criterion has been examined in the past, and the solution obtained, based on the Euler-Poisson equation, failed to predict that the hand would begin and end the movement at rest (i.e., with zero acceleration). Therefore, this criterion was rejected in favor of the minimum jerk, which was proved to be successful in describing many features of human movements. This letter follows an alternative approach and solves the minimum acceleration problem with constraints using Pontryagin's minimum principle. We use the minimum principle to obtain minimum acceleration trajectories and use the jerk as a control signal. In order to find a solution that does not include nonphysiological impulse functions, constraints on the maximum and minimum jerk values are assumed. The analytical solution provides a three-phase piecewise constant jerk signal (bang-bang control) where the magnitude of the jerk and the two switching times depend on the magnitude of the maximum and minimum available jerk values. This result fits the observed trajectories of reaching movements and takes into account both the extrinsic coordinates and the muscle limitations in a single framework. The minimum acceleration with constraints principle is discussed as a unifying approach for many observations about the neural control of movements.

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63. A Generalized Divergence Measure for Nonnegative Matrix Factorization

Raul Kompass

https://doi.org/10.1162/neco.2007.19.3.780 | pp. 780-791

This letter presents a general parametric divergence measure. The metric includes as special cases quadratic error and Kullback-Leibler divergence. A parametric generalization of the two different multiplicative update rules for nonnegative matrix factorization by Lee and Seung (2001) is shown to lead to locally optimal solutions of the nonnegative matrix factorization problem with this new cost function. Numeric simulations demonstrate that the new update rule may improve the quadratic distance convergence speed. A proof of convergence is given that, as in Lee and Seung, uses an auxiliary function known from the expectation-maximization theoretical framework.

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64. Support Vector Ordinal Regression

Wei Chu, S. Sathiya Keerthi
https://doi.org/10.1162/neco.2007.19.3.792 | pp. 792-815

In this letter, we propose two new support vector approaches for ordinal regression, which optimize multiple thresholds to define parallel discriminant hyperplanes for the ordinal scales. Both approaches guarantee that the thresholds are properly ordered at the optimal solution. The size of these optimization problems is linear in the number of training samples. The sequential minimal optimization algorithm is adapted for the resulting optimization problems; it is extremely easy to implement and scales efficiently as a quadratic function of the number of examples. The results of numerical experiments on some benchmark and real-world data sets, including applications of ordinal regression to information retrieval, verify the usefulness of these approaches.

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65. Dynamics of Learning Near Singularities in Layered Networks

Haikun Wei, Jun Zhang, Florent Cousseau, Tomoko Ozeki, ■■■-■■■ ■■■■■
https://doi.org/10.1162/neco.2007.12-06-414 | pp. 813-843

We explicitly analyze the trajectories of learning near singularities in hierarchical networks, such as multilayer perceptrons and radial basis function networks, which include permutation symmetry of hidden nodes, and show their general properties. Such symmetry induces singularities in their parameter space, where the Fisher information matrix degenerates and odd learning behaviors, especially the existence of plateaus in gradient descent learning, arise due to the geometric structure of singularity. We plot dynamic vector fields to demonstrate the universal trajectories of learning near singularities. The singularity induces two types of plateaus, the on-singularity plateau and the near-singularity plateau, depending on the stability of the singularity and the initial parameters of learning. The results presented in this letter are universally applicable to a wide class of hierarchical models. Detailed stability analysis of the dynamics of learning in radial basis function networks and multilayer perceptrons will be presented in separate work.

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66. Neighborhood Property–Based Pattern Selection for Support Vector Machines

Hyunjung Shin, Sungzoon Cho
https://doi.org/10.1162/neco.2007.19.3.816 | pp. 816-855

The support vector machine (SVM) has been spotlighted in the machine learning community because of its theoretical soundness and practical performance. When applied to a large data set, however, it requires a large memory and a long time for training. To cope with the practical difficulty, we propose a pattern selection algorithm based on neighborhood properties. The idea is to select only the patterns that are likely to be located near the decision boundary. Those patterns are expected to be more informative than the randomly selected patterns. The experimental results provide promising evidence that it is possible to successfully employ the proposed algorithm ahead of SVM training.

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67. A Cooperative Recurrent Neural Network for Solving L₁ Estimation Problems with General Linear Constraints

Youshen Xia, Mohamed S. Kamel

https://doi.org/10.1162/neco.2007.10-06-376 | pp. 844-872

The constrained L(1) estimation is an attractive alternative to both the unconstrained L(1) estimation and the least square estimation. In this letter, we propose a cooperative recurrent neural network (CRNN) for solving L(1) estimation problems with general linear constraints. The proposed CRNN model combines four individual neural network models automatically and is suitable for parallel implementation. As a special case, the proposed CRNN includes two existing neural networks for solving unconstrained and constrained L(1) estimation problems, respectively. Unlike existing neural networks, with penalty parameters, for solving the constrained L(1) estimation problem, the proposed CRNN is guaranteed to converge globally to the exact optimal solution without any additional condition. Compared with conventional numerical algorithms, the proposed CRNN has a low computational complexity and can deal with the L(1) estimation problem with degeneracy. Several applied examples show that the proposed CRNN can obtain more accurate estimates than several existing algorithms.

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68. Transductive Methods for the Distributed Ensemble Classification Problem

David J. Miller, Siddharth Pal

https://doi.org/10.1162/neco.2007.19.3.856 | pp. 856-884

We consider ensemble classification for the case where there is no common labeled training data for jointly designing the individual classifiers and the function that aggregates their decisions. This problem, which we call distributed ensemble classification, applies when individual classifiers operate (perhaps remotely) on different sensing modalities and when combining proprietary or legacy classifiers. The conventional wisdom in this case is to apply fixed rules of combination such as voting methods or rules for aggregating probabilities. Alternatively, we take a transductive approach, optimizing the combining rule for an objective function measured on the unlabeled batch of test data. We propose maximum likelihood (ML) objectives that are shown to yield well-known forms of probabilistic aggregation, albeit with iterative, expectation-maximization-based adjustment to account for mismatch between class priors used by individual classifiers and those reflected in the new data batch. These methods are extensions, for the ensemble case, of the work of Saerens, Latinne, and Decaestecker (2002). We also propose an information-theoretic method that generally outperforms the ML methods, better handles classifier redundancies, and addresses some scenarios where the ML methods are not applicable. This method also well handles the case of missing classes in the test batch. On UC Irvine benchmark data, all our methods give improvements in classification accuracy over the use of fixed rules when there is prior mismatch.

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69. The Diffusion Decision Model: Theory and Data for Two-Choice Decision Tasks

Roger Ratcliff, Gail McKoon

https://doi.org/10.1162/neco.2008.12-06-420 | pp. 873-922

The diffusion decision model allows detailed explanations of behavior in two-choice discrimination tasks. In this article, the model is reviewed to show how it translates behavioral data-accuracy, mean response times,

and response time distributions-into components of cognitive processing. Three experiments are used to illustrate experimental manipulations of three components: stimulus difficulty affects the quality of information on which a decision is based; instructions emphasizing either speed or accuracy affect the criterial amounts of information that a subject requires before initiating a response; and the relative proportions of the two stimuli affect biases in drift rate and starting point. The experiments also illustrate the strong constraints that ensure the model is empirically testable and potentially falsifiable. The broad range of applications of the model is also reviewed, including research in the domains of aging and neurophysiology.

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70. Synergies Between Intrinsic and Synaptic Plasticity Mechanisms

Jochen Triesch

https://doi.org/10.1162/neco.2007.19.4.885 | pp. 885-909

We propose a model of intrinsic plasticity for a continuous activation model neuron based on information theory. We then show how intrinsic and synaptic plasticity mechanisms interact and allow the neuron to discover heavy-tailed directions in the input. We also demonstrate that intrinsic plasticity may be an alternative explanation for the sliding threshold postulated in the BCM theory of synaptic plasticity. We present a theoretical analysis of the interaction of intrinsic plasticity with different Hebbian learning rules for the case of clustered inputs. Finally, we perform experiments on the "bars" problem, a popular nonlinear independent component analysis problem.

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71. Distinguishing Causal Interactions in Neural Populations

Anil K. Seth, Gerald M. Edelman

https://doi.org/10.1162/neco.2007.19.4.910 | pp. 910-933

We describe a theoretical network analysis that can distinguish statistically causal interactions in population neural activity leading to a specific output. We introduce the concept of a causal core to refer to the set of neuronal interactions that are causally significant for the output, as assessed by Granger causality. Because our approach requires extensive knowledge of neuronal connectivity and dynamics, an illustrative example is provided by analysis of Darwin X, a brain-based device that allows precise recording of the activity of neuronal units during behavior. In Darwin X, a simulated neuronal model of the hippocampus and surrounding cortical areas supports learning of a spatial navigation task in a real environment. Analysis of Darwin X reveals that large repertoires of neuronal interactions contain comparatively small causal cores and that these causal cores become smaller during learning, a finding that may reflect the selection of specific causal pathways from diverse neuronal repertoires.

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72. Spike Train Decoding Without Spike Sorting

Valérie Ventura

https://doi.org/10.1162/neco.2008.02-07-478 | pp. 923-963

We propose a novel paradigm for spike train decoding, which avoids entirely spike sorting based on waveform measurements. This paradigm directly uses the spike train collected at recording electrodes from thresholding the bandpassed voltage signal. Our approach is a paradigm, not an algorithm, since it can be

used with any of the current decoding algorithms, such as population vector or likelihood-based algorithms. Based on analytical results and an extensive simulation study, we show that our paradigm is comparable to, and sometimes more efficient than, the traditional approach based on well-isolated neurons and that it remains efficient even when all electrodes are severely corrupted by noise, a situation that would render spike sorting particularly difficult. Our paradigm will also save time and computational effort, both of which are crucially important for successful operation of real-time brain-machine interfaces. Indeed, in place of the lengthy spike-sorting task of the traditional approach, it involves an exact expectation EM algorithm that is fast enough that it could also be left to run during decoding to capture potential slow changes in the states of the neurons.

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73. Model Selection for Convolutive ICA with an Application to Spatiotemporal Analysis of EEG

Mads Dyrholm, Scott Makeig, Lars Kai Hansen

https://doi.org/10.1162/neco.2007.19.4.934 | pp. 934-955

We present a new algorithm for maximum likelihood convolutive independent component analysis (ICA) in which components are unmixed using stable autoregressive filters determined implicitly by estimating a convolutive model of the mixing process. By introducing a convolutive mixing model for the components, we show how the order of the filters in the model can be correctly detected using Bayesian model selection. We demonstrate a framework for deconvolving a subspace of independent components in electroencephalography (EEG). Initial results suggest that in some cases, convolutive mixing may be a more realistic model for EEG signals than the instantaneous ICA model.

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74. Information and Topology in Attractor Neural Networks

David Domínguez, Kostadin Koroutchev, Eduardo Serrano, Francisco B. Rodr■■guez

https://doi.org/10.1162/neco.2007.19.4.956 | pp. 956-973

A wide range of networks, including those with small-world topology, can be modeled by the connectivity ratio and randomness of the links. Both learning and attractor abilities of a neural network can be measured by the mutual information (MI) as a function of the load and the overlap between patterns and retrieval states. In this letter, we use MI to search for the optimal topology with regard to the storage and attractor properties of the network in an Amari-Hopfield model. We find that while an optimal storage implies an extremely diluted topology, a large basin of attraction leads to moderate levels of connectivity. This optimal topology is related to the clustering and path length of the network. We also build a diagram for the dynamical phases with random or local initial overlap and show that very diluted networks lose their attractor ability.

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75. Constrained Subspace ICA Based on Mutual Information Optimization Directly

Marc M. Van Hulle

https://doi.org/10.1162/neco.2008.10-06-383 | pp. 964-973

We introduce a new approach to constrained independent component analysis (ICA) by formulating the original, unconstrained ICA problem as well as the constraints in mutual information terms directly. As an estimate of mutual information, a robust version of the Edgeworth expansion is used, on which gradient descent is performed. As an application, we consider the extraction of both the mother and the fetal antepartum electrocardiograms (ECG) from multielectrode cutaneous recordings on the mother's thorax and abdomen.

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76. Dynamical Constraints on Using Precise Spike Timing to Compute in Recurrent Cortical Networks

Arunava Banerjee, Peggy Seriès, Alexandre Pouget

https://doi.org/10.1162/neco.2008.05-06-206 | pp. 974-993

Several recent models have proposed the use of precise timing of spikes for cortical computation. Such models rely on growing experimental evidence that neurons in the thalamus as well as many primary sensory cortical areas respond to stimuli with remarkable temporal precision. Models of computation based on spike timing, where the output of the network is a function not only of the input but also of an independently initializable internal state of the network, must, however, satisfy a critical constraint: the dynamics of the network should not be sensitive to initial conditions. We have previously developed an abstract dynamical system for networks of spiking neurons that has allowed us to identify the criterion for the stationary dynamics of a network to be sensitive to initial conditions. Guided by this criterion, we analyzed the dynamics of several recurrent cortical architectures, including one from the orientation selectivity literature. Based on the results, we conclude that under conditions of sustained, Poisson-like, weakly correlated, low to moderate levels of internal activity as found in the cortex, it is unlikely that recurrent cortical networks can robustly generate precise spike trajectories, that is, spatiotemporal patterns of spikes precise to the millisecond timescale.

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77. Connection Topology Selection in Central Pattern Generators by Maximizing the Gain of Information

Gregory R. Stiesberg, Marcelo Bussotti Reyes, Pablo Varona, Reynaldo D. Pinto, Ramón Huerta

https://doi.org/10.1162/neco.2007.19.4.974 | pp. 974-993

A study of a general central pattern generator (CPG) is carried out by means of a measure of the gain of information between the number of available topology configurations and the output rhythmic activity. The neurons of the CPG are chaotic Hindmarsh-Rose models that cooperate dynamically to generate either chaotic or regular spatiotemporal patterns. These model neurons are implemented by computer simulations and electronic circuits. Out of a random pool of input configurations, a small subset of them maximizes the gain of information. Two important characteristics of this subset are emphasized: (1) the most regular output activities are chosen, and (2) none of the selected input configurations are networks with open topology. These two principles are observed in living CPGs as well as in model CPGs that are the most efficient in controlling mechanical tasks, and they are evidence that the information-theoretical analysis can be an invaluable tool in searching for general properties of CPGs.

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78. Dynamics and Computation of Continuous Attractors

Si Wu, Kosuke Hamaguchi, ■■■-■■■ ■■■■■

https://doi.org/10.1162/neco.2008.10-06-378 | pp. 994-1025

Continuous attractor is a promising model for describing the encoding of continuous stimuli in neural systems. In a continuous attractor, the stationary states of the neural system form a continuous parameter space, on which the system is neutrally stable. This property enables the neutral system to track time-varying stimuli smoothly, but it also degrades the accuracy of information retrieval, since these stationary states are easily disturbed by external noise. In this work, based on a simple model, we systematically investigate the dynamics and the computational properties of continuous attractors. In order to analyze the dynamics of a large-size network, which is otherwise extremely complicated, we develop a strategy to reduce its dimensionality by utilizing the fact that a continuous attractor can eliminate the noise components perpendicular to the attractor space very quickly. We therefore project the network dynamics onto the tangent of the attractor space and simplify it successfully as a one-dimensional Ornstein-Uhlenbeck process. Based on this simplified model, we investigate (1) the decoding error of a continuous attractor under the driving of external noisy inputs, (2) the tracking speed of a continuous attractor when external stimulus experiences abrupt changes, (3) the neural correlation structure associated with the specific dynamics of a continuous attractor, and (4) the consequence of asymmetric neural correlation on statistical population decoding. The potential implications of these results on our understanding of neural information processing are also discussed.

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79. Independent Slow Feature Analysis and Nonlinear Blind Source Separation

Tobias Blaschke, Tiziano Zito, Laurenz Wiskott

https://doi.org/10.1162/neco.2007.19.4.994 | pp. 994-1021

In the linear case, statistical independence is a sufficient criterion for performing blind source separation. In the nonlinear case, however, it leaves an ambiguity in the solutions that has to be resolved by additional criteria. Here we argue that temporal slowness complements statistical independence well and that a combination of the two leads to unique solutions of the nonlinear blind source separation problem. The algorithm we present is a combination of second-order independent component analysis and slow feature analysis and is referred to as independent slow feature analysis. Its performance is demonstrated on nonlinearly mixed music data. We conclude that slowness is indeed a useful complement to statistical independence but that time-delayed second-order moments are only a weak measure of statistical independence.

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80. A Maximum-Likelihood Interpretation for Slow Feature Analysis

Richard M. Turner, Maneesh Sahani

https://doi.org/10.1162/neco.2007.19.4.1022 | pp. 1022-1038

The brain extracts useful features from a maelstrom of sensory information, and a fundamental goal of theoretical neuroscience is to work out how it does so. One proposed feature extraction strategy is motivated by the observation that the meaning of sensory data, such as the identity of a moving visual

object, is often more persistent than the activation of any single sensory receptor. This notion is embodied in the slow feature analysis (SFA) algorithm, which uses "slowness" as a heuristic by which to extract semantic information from multidimensional time series. Here, we develop a probabilistic interpretation of this algorithm, showing that inference and learning in the limiting case of a suitable probabilistic model yield exactly the results of SFA. Similar equivalences have proved useful in interpreting and extending comparable algorithms such as independent component analysis. For SFA, we use the equivalent probabilistic model as a conceptual springboard with which to motivate several novel extensions to the algorithm.

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81. Predictive Coding and the Slowness Principle: An Information-Theoretic Approach

Felix Creutzig, Henning Sprekeler

https://doi.org/10.1162/neco.2008.01-07-455 | pp. 1026-1041

Understanding the guiding principles of sensory coding strategies is a main goal in computational neuroscience. Among others, the principles of predictive coding and slowness appear to capture aspects of sensory processing. Predictive coding postulates that sensory systems are adapted to the structure of their input signals such that information about future inputs is encoded. Slow feature analysis (SFA) is a method for extracting slowly varying components from quickly varying input signals, thereby learning temporally invariant features. Here, we use the information bottleneck method to state an information-theoretic objective function for temporally local predictive coding. We then show that the linear case of SFA can be interpreted as a variant of predictive coding that maximizes the mutual information between the current output of the system and the input signal in the next time step. This demonstrates that the slowness principle and predictive coding are intimately related.

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82. An Augmented Extended Kalman Filter Algorithm for Complex-Valued Recurrent Neural Networks

Su Lee Goh, Danilo P. Mandic

https://doi.org/10.1162/neco.2007.19.4.1039 | pp. 1039-1055

An augmented complex-valued extended Kalman filter (ACEKF) algorithm for the class of nonlinear adaptive filters realized as fully connected recurrent neural networks is introduced. This is achieved based on some recent developments in the so-called augmented complex statistics and the use of general fully complex nonlinear activation functions within the neurons. This makes the ACEKF suitable for processing general complex-valued nonlinear and nonstationary signals and also bivariate signals with strong component correlations. Simulations on benchmark and real-world complex-valued signals support the approach.

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83. A Homomorphic Neural Network for Modeling and Prediction

Maciej P■dzisz, Danilo P. Mandic

https://doi.org/10.1162/neco.2008.12-06-418 | pp. 1042-1064

A homomorphic feedforward network (HFFN) for nonlinear adaptive filtering is introduced. This is achieved by a two-layer feedforward architecture with an exponential hidden layer and logarithmic preprocessing step. This way, the overall input-output relationship can be seen as a generalized Volterra model, or as a bank of homomorphic filters. Gradient-based learning for this architecture is introduced, together with some practical issues related to the choice of optimal learning parameters and weight initialization. The performance and convergence speed are verified by analysis and extensive simulations. For rigor, the simulations are conducted on artificial and real-life data, and the performances are compared against those obtained by a sigmoidal feedforward network (FFN) with identical topology. The proposed HFFN proved to be a viable alternative to FFNs, especially in the critical case of online learning on small- and medium-scale data sets.

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84. Equilibria of Iterative Softmax and Critical Temperatures for Intermittent Search in Self-Organizing Neural Networks

Peter Ti■o

https://doi.org/10.1162/neco.2007.19.4.1056 | pp. 1056-1081

Kwok and Smith (2005) recently proposed a new kind of optimization dynamics using self-organizing neural networks (SONN) driven by softmax weight renormalization. Such dynamics is capable of powerful intermittent search for high-quality solutions in difficult assignment optimization problems. However, the search is sensitive to temperature setting in the softmax renormalization step. It has been hypothesized that the optimal temperature setting corresponds to the symmetry-breaking bifurcation of equilibria of the renormalization step, when viewed as an autonomous dynamical system called iterative softmax (ISM). We rigorously analyze equilibria of ISM by determining their number, position, and stability types. It is shown that most fixed points exist in the neighborhood of the maximum entropy equilibrium w= (N(-1), N(-1), ..., N(-1)), where N is the ISM dimensionality. We calculate the exact rate of decrease in the number of ISM equilibria as one moves away from w. Bounds on temperatures guaranteeing different stability types of ISM equilibria are also derived. Moreover, we offer analytical approximations to the critical symmetry-breaking bifurcation temperatures that are in good agreement with those found by numerical investigations. So far, the critical temperatures have been determined only by trial-and-error numerical simulations. On a set of N-queens problems for a wide range of problem sizes N, the analytically determined critical temperatures predict the optimal working temperatures for SONN intermittent search very well. It is also shown that no intermittent search can exist in SONN for temperatures greater than one-half.

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85. Almost Periodic Dynamics of a Class of Delayed Neural Networks with Discontinuous Activations

Wenlian Lu, Tianping Chen

https://doi.org/10.1162/neco.2008.10-06-364 | pp. 1065-1090

We use the concept of the Filippov solution to study the dynamics of a class of delayed dynamical systems with discontinuous right-hand side, which contains the widely studied delayed neural network models with almost periodic self-inhibitions, interconnection weights, and external inputs. We prove that diagonal-dominant conditions can guarantee the existence and uniqueness of an almost periodic solution, as well as its global exponential stability. As special cases, we derive a series of results on the dynamics of delayed dynamical systems with discontinuous activations and periodic coefficients or constant coefficients,

respectively. From the proof of the existence and uniqueness of the solution, we prove that the solution of a delayed dynamical system with high-slope activations approximates to the Filippov solution of the dynamical system with discontinuous activations.

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86. Recursive Finite Newton Algorithm for Support Vector Regression in the Primal

Liefeng Bo, Ling Wang, Licheng Jiao

https://doi.org/10.1162/neco.2007.19.4.1082 | pp. 1082-1096

Some algorithms in the primal have been recently proposed for training support vector machines. This letter follows those studies and develops a recursive finite Newton algorithm (IHLF-SVR-RFN) for training nonlinear support vector regression. The insensitive Huber loss function and the computation of the Newton step are discussed in detail. Comparisons with LIBSVM 2.82 show that the proposed algorithm gives promising results.

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87. A Study on Neural Learning on Manifold Foliations: The Case of the Lie Group SU(3)

Simone Fiori

https://doi.org/10.1162/neco.2008.03-07-489 | pp. 1091-1117

Learning on differential manifolds may involve the optimization of a function of many parameters. In this letter, we deal with Riemannian-gradient-based optimization on a Lie group, namely, the group of unitary unimodular matrices SU(3). In this special case, subalgebras of the associated Lie algebra su(3) may be individuated by computing pair-wise Gell-Mann matrices commutators. Subalgebras generate subgroups of a Lie group, as well as manifold foliation. We show that the Riemannian gradient may be projected over tangent structures to foliation, giving rise to foliation gradients. Exponentiations of foliation gradients may be computed in closed forms, which closely resemble Rodriguez forms for the special orthogonal group SO(3). We thus compare optimization by Riemannian gradient and foliation gradients.

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88. State-Space Models: From the EM Algorithm to a Gradient Approach

Rasmus Kongsgaard Olsson, Kaare Brandt Petersen, Tue Lehn-Schiøler

https://doi.org/10.1162/neco.2007.19.4.1097 | pp. 1097-1111

Slow convergence is observed in the EM algorithm for linear state-space models. We propose to circumvent the problem by applying any off-the-shelf quasi-Newton-type optimizer, which operates on the gradient of the log-likelihood function. Such an algorithm is a practical alternative due to the fact that the exact gradient of the log-likelihood function can be computed by recycling components of the expectation-maximization (EM) algorithm. We demonstrate the efficiency of the proposed method in three relevant instances of the linear state-space model. In high signal-to-noise ratios, where EM is particularly prone to converge slowly, we show that gradient-based learning results in a sizable reduction of computation time.

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89. Variational Bayes Solution of Linear Neural Networks and Its Generalization Performance

Shinichi Nakajima, Sumio Watanabe

https://doi.org/10.1162/neco.2007.19.4.1112 | pp. 1112-1153

It is well known that in unidentifiable models, the Bayes estimation provides much better generalization performance than the maximum likelihood (ML) estimation. However, its accurate approximation by Markov chain Monte Carlo methods requires huge computational costs. As an alternative, a tractable approximation method, called the variational Bayes (VB) approach, has recently been proposed and has been attracting attention. Its advantage over the expectation maximization (EM) algorithm, often used for realizing the ML estimation, has been experimentally shown in many applications; nevertheless, it has not yet been theoretically shown. In this letter, through analysis of the simplest unidentifiable models, we theoretically show some properties of the VB approach. We first prove that in three-layer linear neural networks, the VB approach is asymptotically equivalent to a positive-part James-Stein type shrinkage estimation. Then we theoretically clarify its free energy, generalization error, and training error. Comparing them with those of the ML estimation and the Bayes estimation, we discuss the advantage of the VB approach. We also show that unlike in the Bayes estimation, the free energy and the generalization error are less simply related with each other and that in typical cases, the VB free energy well approximates the Bayes one, while the VB generalization error significantly differs from the Bayes one.

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90. Training a Support Vector Machine in the Primal

Olivier Chapelle

https://doi.org/10.1162/neco.2007.19.5.1155 | pp. 1155-1178

Most literature on support vector machines (SVMs) concentrates on the dual optimization problem. In this letter, we point out that the primal problem can also be solved efficiently for both linear and nonlinear SVMs and that there is no reason for ignoring this possibility. On the contrary, from the primal point of view, new families of algorithms for large-scale SVM training can be investigated.

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91. Populations of Tightly Coupled Neurons: The RGC/LGN System

Lawrence Sirovich

https://doi.org/10.1162/neco.2007.03-07-482 | pp. 1179-1210

A mathematical model, of general character for the dynamic description of coupled neural oscillators is presented. The population approach that is employed applies equally to coupled cells as to populations of such coupled cells. The formulation includes stochasticity and preserves details of precisely firing neurons. Based on the generally accepted view of cortical wiring, this formulation is applied to the retinal ganglion cell (RGC)/lateral geniculate nucleus (LGN) relay cell system, of the early mammalian visual system. The smallness of quantal voltage jumps at the retinal level permits a Fokker-Planck approximation for the RGC contribution; however, the LGN description requires the use of finite jumps, which for fast synaptic dynamics appears as finite jumps in the membrane potential. Analyses of equilibrium spiking behavior for both the deterministic and stochastic cases are presented. Green's function methods form the basis for the asymptotic and exact results that are presented. This determines the spiking ratio (i.e., the number of RGC

arrivals per LGN spike), which is the reciprocal of the transfer ratio, under wide circumstances. Criteria for spiking regimes, in terms of the relatively few parameters of the model, are presented. Under reasonable hypotheses, it is shown that the transfer ratio is <or=1/2, in the absence of input from other areas. Thus, the model suggests that the LGN/RGC system may be a relatively unsophisticated spike editor. In the absence of other input, the system is designed to fire an LGN spike only when two or more RGC spikes appear in a relatively short time. Transfer ratios that briefly exceed 1/2 (but are less than 1) have been recorded in the laboratory. Inclusion of brain stem input has been shown to provide a signal that elevates the transfer ratio (Ozaki & Kaplan, 2006). A model that includes this contribution is also presented.

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92. Dynamics of Nonlinear Feedback Control

H. P. Snippe, J. H. van Hateren

https://doi.org/10.1162/neco.2007.19.5.1179 | pp. 1179-1214

Feedback control in neural systems is ubiquitous. Here we study the mathematics of nonlinear feedback control. We compare models in which the input is multiplied by a dynamic gain (multiplicative control) with models in which the input is divided by a dynamic attenuation (divisive control). The gain signal (resp. the attenuation signal) is obtained through a concatenation of an instantaneous nonlinearity and a linear low-pass filter operating on the output of the feedback loop. For input steps, the dynamics of gain and attenuation can be very different, depending on the mathematical form of the nonlinearity and the ordering of the nonlinearity and the filtering in the feedback loop. Further, the dynamics of feedback control can be strongly asymmetrical for increment versus decrement steps of the input. Nevertheless, for each of the models studied, the nonlinearity in the feedback loop can be chosen such that immediately after an input step, the dynamics of feedback control is symmetric with respect to increments versus decrements. Finally, we study the dynamics of the output of the control loops and find conditions under which overshoots and undershoots of the output relative to the steady-state output occur when the models are stimulated with low-pass filtered steps. For small steps at the input, overshoots and undershoots of the output do not occur when the filtering in the control path is faster than the low-pass filtering at the input. For large steps at the input, however, results depend on the model, and for some of the models, multiple overshoots and undershoots can occur even with a fast control path.

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93. Interspike Interval Statistics in the Stochastic Hodgkin-Huxley Model: Coexistence of Gamma Frequency Bursts and Highly Irregular Firing

Peter F. Rowat

https://doi.org/10.1162/neco.2007.19.5.1215 | pp. 1215-1250

When the classical Hodgkin-Huxley equations are simulated with Na- and K-channel noise and constant applied current, the distribution of interspike intervals is bimodal: one part is an exponential tail, as often assumed, while the other is a narrow gaussian peak centered at a short interspike interval value. The gaussian arises from bursts of spikes in the gamma-frequency range, the tail from the interburst intervals, giving overall an extraordinarily high coefficient of variation--up to 2.5 for 180,000 Na channels when I approximately 7 microA/cm(2). Since neurons with a bimodal ISI distribution are common, it may be a useful model for any neuron with class 2 firing. The underlying mechanism is due to a subcritical Hopf bifurcation, together with a switching region in phase-space where a fixed point is very close to a system limit cycle. This

mechanism may be present in many different classes of neurons and may contribute to widely observed highly irregular neural spiking.

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94. The Firing of an Excitable Neuron in the Presence of Stochastic Trains of Strong Synaptic Inputs

Jonathan E. Rubin, Krešimir Josi■■

https://doi.org/10.1162/neco.2007.19.5.1251 | pp. 1251-1294

We consider a fast-slow excitable system subject to a stochastic excitatory input train and show that under general conditions, its long-term behavior is captured by an irreducible Markov chain with a limiting distribution. This limiting distribution allows for the analytical calculation of the system's probability of firing in response to each input, the expected number of response failures between firings, and the distribution of slow variable values between firings. Moreover, using this approach, it is possible to understand why the system will not have a stationary distribution and why Monte Carlo simulations do not converge under certain conditions. The analytical calculations involved can be performed whenever the distribution of interexcitation intervals and the recovery dynamics of the slow variable are known. The method can be extended to other models that feature a single variable that builds up to a threshold where an instantaneous spike and reset occur. We also discuss how the Markov chain analysis generalizes to any pair of input trains, excitatory or inhibitory and synaptic or not, such that the frequencies of the two trains are sufficiently different from each other. We illustrate this analysis on a model thalamocortical (TC) cell subject to two example distributions of excitatory synaptic inputs in the cases of constant and rhythmic inhibition. The analysis shows a drastic drop in the likelihood of firing just after inhibitory onset in the case of rhythmic inhibition, relative even to the case of elevated but constant inhibition. This observation provides support for a possible mechanism for the induction of motor symptoms in Parkinson's disease and for their relief by deep brain stimulation, analyzed in Rubin and Terman (2004).

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95. Phase Precession Through Synaptic Facilitation

Kay Thurley, Christian Leibold, Anja Gundlfinger, Dietmar Schmitz, Richard Kempter

https://doi.org/10.1162/neco.2008.07-06-292 | pp. 1285-1324

Phase precession is a relational code that is thought to be important for episodic-like memory, for instance, the learning of a sequence of places. In the hippocampus, places are encoded through bursting activity of so-called place cells. The spikes in such a burst exhibit a precession of their firing phases relative to field potential theta oscillations (4-12 Hz); the theta phase of action potentials in successive theta cycles progressively decreases toward earlier phases. The mechanisms underlying the generation of phase precession are, however, unknown. In this letter, we show through mathematical analysis and numerical simulations that synaptic facilitation in combination with membrane potential oscillations of a neuron gives rise to phase precession. This biologically plausible model reproduces experimentally observed features of phase precession, such as (1) the progressive decrease of spike phases, (2) the nonlinear and often also bimodal relation between spike phases and the animal's place, (3) the range of phase precession being smaller than one theta cycle, and (4) the dependence of phase jitter on the animal's location within the place field. The model suggests that the peculiar features of the hippocampal mossy fiber synapse, such as its large efficacy, long-lasting and strong facilitation, and its phase-locked activation, are essential for phase

precession in the CA3 region of the hippocampus.

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96. Optimal Coding Predicts Attentional Modulation of Activity in Neural Systems

Santiago Jaramillo, Barak A. Pearlmutter

https://doi.org/10.1162/neco.2007.19.5.1295 | pp. 1295-1312

Neuronal activity in response to a fixed stimulus has been shown to change as a function of attentional state, implying that the neural code also changes with attention. We propose an information-theoretic account of such modulation: that the nervous system adapts to optimally encode sensory stimuli while taking into account the changing relevance of different features. We show using computer simulation that such modulation emerges in a coding system informed about the uneven relevance of the input features. We present a simple feedforward model that learns a covert attention mechanism, given input patterns and coding fidelity requirements. After optimization, the system gains the ability to reorganize its computational resources (and coding strategy) depending on the incoming attentional signal, without the need of multiplicative interaction or explicit gating mechanisms between units. The modulation of activity for different attentional states matches that observed in a variety of selective attention experiments. This model predicts that the shape of the attentional modulation function can be strongly stimulus dependent. The general principle presented here accounts for attentional modulation of neural activity without relying on special-purpose architectural mechanisms dedicated to attention. This principle applies to different attentional goals, and its implications are relevant for all modalities in which attentional phenomena are observed.

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97. Efficient Computation Based on Stochastic Spikes

Udo Ernst, David Rotermund, Klaus Pawelzik

https://doi.org/10.1162/neco.2007.19.5.1313 | pp. 1313-1343

The speed and reliability of mammalian perception indicate that cortical computations can rely on very few action potentials per involved neuron. Together with the stochasticity of single-spike events in cortex, this appears to imply that large populations of redundant neurons are needed for rapid computations with action potentials. Here we demonstrate that very fast and precise computations can be realized also in small networks of stochastically spiking neurons. We present a generative network model for which we derive biologically plausible algorithms that perform spike-by-spike updates of the neuron's internal states and adaptation of its synaptic weights from maximizing the likelihood of the observed spike patterns. Paradigmatic computational tasks demonstrate the online performance and learning efficiency of our framework. The potential relevance of our approach as a model for cortical computation is discussed.

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98. Exact Inferences in a Neural Implementation of a Hidden Markov Model

Jeffrey M. Beck, Alexandre Pouget

https://doi.org/10.1162/neco.2007.19.5.1344 | pp. 1344-1361

From first principles, we derive a quadratic nonlinear, first-order dynamical system capable of performing exact Bayes-Markov inferences for a wide class of biologically plausible stimulus-dependent patterns of activity while simultaneously providing an online estimate of model performance. This is accomplished by constructing a dynamical system that has solutions proportional to the probability distribution over the stimulus space, but with a constant of proportionality adjusted to provide a local estimate of the probability of the recent observations of stimulus-dependent activity-given model parameters. Next, we transform this exact equation to generate nonlinear equations for the exact evolution of log likelihood and log-likelihood ratios and show that when the input has low amplitude, linear rate models for both the likelihood and the log-likelihood functions follow naturally from these equations. We use these four explicit representations of the probability distribution to argue that, in contrast to the arguments of previous work, the dynamical system for the exact evolution of the likelihood (as opposed to the log likelihood or log-likelihood ratios) not only can be mapped onto a biologically plausible network but is also more consistent with physiological observations.

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99. Sequential Fixed-Point ICA Based on Mutual Information Minimization

Marc M. Van Hulle

https://doi.org/10.1162/neco.2008.09-06-335 | pp. 1344-1365

A new gradient technique is introduced for linear independent component analysis (ICA) based on the Edgeworth expansion of mutual information, for which the algorithm operates sequentially using fixed-point iterations. In order to address the adverse effect of outliers, a robust version of the Edgeworth expansion is adopted, in terms of robust cumulants, and robust derivatives of the Hermite polynomials are used. Also, a new constrained version of ICA is introduced, based on goal programming of mutual information objectives, which is applied to the extraction of the antepartum fetal electrocardiogram from multielectrode cutaneous recordings on the mother's thorax and abdomen.

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100. Multispike Interactions in a Stochastic Model of Spike-Timing-Dependent Plasticity

Peter A. Appleby, Terry Elliott

https://doi.org/10.1162/neco.2007.19.5.1362 | pp. 1362-1399

Recently we presented a stochastic, ensemble-based model of spike-timing-dependent plasticity. In this model, single synapses do not exhibit plasticity depending on the exact timing of pre- and postsynaptic spikes, but spike-timing-dependent plasticity emerges only at the temporal or synaptic ensemble level. We showed that such a model reproduces a variety of experimental results in a natural way, without the introduction of various, ad hoc nonlinearities characteristic of some alternative models. Our previous study was restricted to an examination, analytically, of two-spike interactions, while higher-order, multispike interactions were only briefly examined numerically. Here we derive exact, analytical results for the general n-spike interaction functions in our model. Our results form the basis for a detailed examination, performed elsewhere, of the significant differences between these functions and the implications these differences have for the presence, or otherwise, of stable, competitive dynamics in our model.

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101. A One-Layer Recurrent Neural Network with a Discontinuous Activation Function for Linear Programming

Qingshan Liu, Jun Wang

https://doi.org/10.1162/neco.2007.03-07-488 | pp. 1366-1383

A one-layer recurrent neural network with a discontinuous activation function is proposed for linear programming. The number of neurons in the neural network is equal to that of decision variables in the linear programming problem. It is proven that the neural network with a sufficiently high gain is globally convergent to the optimal solution. Its application to linear assignment is discussed to demonstrate the utility of the neural network. Several simulation examples are given to show the effectiveness and characteristics of the neural network.

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102. 3D Periodic Human Motion Reconstruction from 2D Motion Sequences

Zonghua Zhang, Nikolaus F. Troje

https://doi.org/10.1162/neco.2007.19.5.1400 | pp. 1400-1421

We present and evaluate a method of reconstructing three-dimensional (3D) periodic human motion from two-dimensional (2D) motion sequences. Using Fourier decomposition, we construct a compact representation for periodic human motion. A low-dimensional linear motion model is learned from a training set of 3D Fourier representations by means of principal components analysis. Two-dimensional test data are projected onto this model with two approaches: least-square minimization and calculation of a maximum a posteriori probability using the Bayes' rule. We present two different experiments in which both approaches are applied to 2D data obtained from 3D walking sequences projected onto a plane. In the first experiment, we assume the viewpoint is known. In the second experiment, the horizontal viewpoint is unknown and is recovered from the 2D motion data. The results demonstrate that by using the linear model, not only can missing motion data be reconstructed, but unknown view angles for 2D test data can also be retrieved.

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103. A New Backpropagation Learning Algorithm for Layered Neural Networks with Nondifferentiable Units

Takahumi Oohori, Hidenori Naganuma, Kazuhisa Watanabe

https://doi.org/10.1162/neco.2007.19.5.1422 | pp. 1422-1435

We propose a digital version of the backpropagation algorithm (DBP) for three-layered neural networks with nondifferentiable binary units. This approach feeds teacher signals to both the middle and output layers, whereas with a simple perceptron, they are given only to the output layer. The additional teacher signals enable the DBP to update the coupling weights not only between the middle and output layers but also between the input and middle layers. A neural network based on DBP learning is fast and easy to implement in hardware. Simulation results for several linearly nonseparable problems such as XOR demonstrate that the DBP performs favorably when compared to the conventional approaches. Furthermore, in large-scale networks, simulation results indicate that the DBP provides high performance.

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104. Spike-Timing-Dependent Plasticity in Balanced Random Networks

Abigail Morrison, Ad Aertsen, Markus Diesmann
https://doi.org/10.1162/neco.2007.19.6.1437 | pp. 1437-1467

The balanced random network model attracts considerable interest because it explains the irregular spiking activity at low rates and large membrane potential fluctuations exhibited by cortical neurons in vivo. In this article, we investigate to what extent this model is also compatible with the experimentally observed phenomenon of spike-timing-dependent plasticity (STDP). Confronted with the plethora of theoretical models for STDP available, we reexamine the experimental data. On this basis, we propose a novel STDP update rule, with a multiplicative dependence on the synaptic weight for depression, and a power law dependence for potentiation. We show that this rule, when implemented in large, balanced networks of realistic connectivity and sparseness, is compatible with the asynchronous irregular activity regime. The resultant equilibrium weight distribution is unimodal with fluctuating individual weight trajectories and does not exhibit development of structure. We investigate the robustness of our results with respect to the relative strength of depression. We introduce synchronous stimulation to a group of neurons and demonstrate that the decoupling of this group from the rest of the network is so severe that it cannot effectively control the spiking of other neurons, even those with the highest convergence from this group.

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105. Reinforcement Learning Through Modulation of Spike-Timing-Dependent Synaptic Plasticity

R■zvan V. Florian
https://doi.org/10.1162/neco.2007.19.6.1468 | pp. 1468-1502

The persistent modification of synaptic efficacy as a function of the relative timing of pre- and postsynaptic spikes is a phenomenon known as spike-timing-dependent plasticity (STDP). Here we show that the modulation of STDP by a global reward signal leads to reinforcement learning. We first derive analytically learning rules involving reward-modulated spike-timing-dependent synaptic and intrinsic plasticity, by applying a reinforcement learning algorithm to the stochastic spike response model of spiking neurons. These rules have several features common to plasticity mechanisms experimentally found in the brain. We then demonstrate in simulations of networks of integrate-and-fire neurons the efficacy of two simple learning rules involving modulated STDP. One rule is a direct extension of the standard STDP model (modulated STDP), and the other one involves an eligibility trace stored at each synapse that keeps a decaying memory of the relationships between the recent pairs of pre- and postsynaptic spike pairs (modulated STDP with eligibility trace). This latter rule permits learning even if the reward signal is delayed. The proposed rules are able to solve the XOR problem with both rate coded and temporally coded input and to learn a target output firing-rate pattern. These learning rules are biologically plausible, may be used for training generic artificial spiking neural networks, regardless of the neural model used, and suggest the experimental investigation in animals of the existence of reward-modulated STDP.

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106. A Method for Selecting the Bin Size of a Time Histogram

Hideaki Shimazaki, Shigeru Shinomoto
https://doi.org/10.1162/neco.2007.19.6.1503 | pp. 1503-1527

The time histogram method is the most basic tool for capturing a time dependent rate of neuronal spikes. Generally in the neurophysiological literature, the bin size that critically determines the goodness of the fit of the time histogram to the underlying spike rate has been subjectively selected by individual researchers. Here, we propose a method for objectively selecting the bin size from the spike count statistics alone, so that the resulting bar or line graph time histogram best represents the unknown underlying spike rate. For a small number of spike sequences generated from a modestly fluctuating rate, the optimal bin size may diverge, indicating that any time histogram is likely to capture a spurious rate. Given a paucity of data, the method presented here can nevertheless suggest how many experimental trials should be added in order to obtain a meaningful time-dependent histogram with the required accuracy.

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107. Penalized Probabilistic Clustering

Zhengdong Lu, Todd K. Leen

https://doi.org/10.1162/neco.2007.19.6.1528 | pp. 1528-1567

While clustering is usually an unsupervised operation, there are circumstances in which we believe (with varying degrees of certainty) that items A and B should be assigned to the same cluster, while items A and C should not. We would like such pairwise relations to influence cluster assignments of out-of-sample data in a manner consistent with the prior knowledge expressed in the training set. Our starting point is probabilistic clustering based on gaussian mixture models (GMM) of the data distribution. We express clustering preferences in a prior distribution over assignments of data points to clusters. This prior penalizes cluster assignments according to the degree with which they violate the preferences. The model parameters are fit with the expectation-maximization (EM) algorithm. Our model provides a flexible framework that encompasses several other semisupervised clustering models as its special cases. Experiments on artificial and real-world problems show that our model can consistently improve clustering results when pairwise relations are incorporated. The experiments also demonstrate the superiority of our model to other semisupervised clustering methods on handling noisy pairwise relations.

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108. Robustness of Connectionist Swimming Controllers Against Random Variation in Neural Connections

Jimmy Or

https://doi.org/10.1162/neco.2007.19.6.1568 | pp. 1568-1588

The ability to achieve high swimming speed and efficiency is very important to both the real lamprey and its robotic implementation. In previous studies, we used evolutionary algorithms to evolve biologically plausible connectionist swimming controllers for a simulated lamprey. This letter investigates the robustness and optimality of the best-evolved controllers as well as the biological controller hand-crafted by Ekeberg. Comparing cases of random variation in intrasegmental or intersegmental weights against each controller allows estimates of robustness to be made. We conduct experiments on the controllers' robustness at the excitation level, which corresponds to either the maximum swimming speed or efficiency by randomly varying the segmental connection weights and on some occasions also the intersegmental couplings, through varying noise ranges. Interestingly, although the swimming performance (in terms of maximum speed and efficiency) of the Ekeberg biological controller is not as good as that of the artificially evolved controllers, it is relatively robust against noise in the neural networks. This suggests that the natural

evolutions have evolved a swimming controller that is good enough to survive in the real world. Our findings could inspire neurobiologists to conduct real physiological experiments to gain a better understanding on how neural connectivity affects behavior. The results can also be applied to control an artificial lamprey in simulation and possibly also a robotic one.

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109. A Measurement Fusion Method for Nonlinear System Identification Using a Cooperative Learning Algorithm

Youshen Xia, Mohamed S. Kamel

https://doi.org/10.1162/neco.2007.19.6.1589 | pp. 1589-1632

Identification of a general nonlinear noisy system viewed as an estimation of a predictor function is studied in this article. A measurement fusion method for the predictor function estimate is proposed. In the proposed scheme, observed data are first fused by using an optimal fusion technique, and then the optimal fused data are incorporated in a nonlinear function estimator based on a robust least squares support vector machine (LS-SVM). A cooperative learning algorithm is proposed to implement the proposed measurement fusion method. Compared with related identification methods, the proposed method can minimize both the approximation error and the noise error. The performance analysis shows that the proposed optimal measurement fusion function estimate has a smaller mean square error than the LS-SVM function estimate. Moreover, the proposed cooperative learning algorithm can converge globally to the optimal measurement fusion function estimate. Finally, the proposed measurement fusion method is applied to ARMA signal and spatial temporal signal modeling. Experimental results show that the proposed measurement fusion method can provide a more accurate model.

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110. Efficient Computation and Model Selection for the Support Vector Regression

Lacey Gunter, Ji Zhu

https://doi.org/10.1162/neco.2007.19.6.1633 | pp. 1633-1655

In this letter, we derive an algorithm that computes the entire solution path of the support vector regression (SVR). We also propose an unbiased estimate for the degrees of freedom of the SVR model, which allows convenient selection of the regularization parameter.

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111. A Novel Generic Hebbian Ordering-Based Fuzzy Rule Base Reduction Approach to Mamdani Neuro-Fuzzy System

Feng Liu, Chai Quek, Geok See Ng

https://doi.org/10.1162/neco.2007.19.6.1656 | pp. 1656-1680

There are two important issues in neuro-fuzzy modeling: (1) interpretability--the ability to describe the behavior of the system in an interpretable way--and (2) accuracy--the ability to approximate the outcome of the system accurately. As these two objectives usually exert contradictory requirements on the neuro-fuzzy model, certain compromise has to be undertaken. This letter proposes a novel rule reduction algorithm, namely, Hebb rule reduction, and an iterative tuning process to balance interpretability and accuracy. The

Hebb rule reduction algorithm uses Hebbian ordering, which represents the degree of coverage of the samples by the rule, as an importance measure of each rule to merge the membership functions and hence reduces the number of the rules. Similar membership functions (MFs) are merged by a specified similarity measure in an order of Hebbian importance, and the resultant equivalent rules are deleted from the rule base. The rule with a higher Hebbian importance will be retained among a set of rules. The MFs are tuned through the least mean square (LMS) algorithm to reduce the modeling error. The tuning of the MFs and the reduction of the rules proceed iteratively to achieve a balance between interpretability and accuracy. Three published data sets by Nakanishi (Nakanishi, Turksen, & Sugeno, 1993), the Pat synthetic data set (Pal, Mitra, & Mitra, 2003), and the traffic flow density prediction data set are used as benchmarks to demonstrate the effectiveness of the proposed method. Good interpretability, as well as high modeling accuracy, are derivable simultaneously and are suitably benchmarked against other well-established neuro-fuzzy models.

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112. Erratum

(authors unknown)

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No abstract available.

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113. Estimating Information Rates with Confidence Intervals in Neural Spike Trains

Jonathon Shlens, Matthew B. Kennel, Henry D. I. Abarbanel, E. J. Chichilnisky

https://doi.org/10.1162/neco.2007.19.7.1683 | pp. 1683-1719

Information theory provides a natural set of statistics to quantify the amount of knowledge a neuron conveys about a stimulus. A related work (Kennel, Shlens, Abarbanel, & Chichilnisky, 2005) demonstrated how to reliably estimate, with a Bayesian confidence interval, the entropy rate from a discrete, observed time series. We extend this method to measure the rate of novel information that a neural spike train encodes about a stimulus--the average and specific mutual information rates. Our estimator makes few assumptions about the underlying neural dynamics, shows excellent performance in experimentally relevant regimes, and uniquely provides confidence intervals bounding the range of information rates compatible with the observed spike train. We validate this estimator with simulations of spike trains and highlight how stimulus parameters affect its convergence in bias and variance. Finally, we apply these ideas to a recording from a guinea pig retinal ganglion cell and compare results to a simple linear decoder.

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114. Generation of Synthetic Spike Trains with Defined Pairwise Correlations

Ernst Niebur

https://doi.org/10.1162/neco.2007.19.7.1720 | pp. 1720-1738

Recent technological advances as well as progress in theoretical understanding of neural systems have created a need for synthetic spike trains with controlled mean rate and pairwise cross-correlation. This report introduces and analyzes a novel algorithm for the generation of discretized spike trains with arbitrary mean rates and controlled cross correlation. Pairs of spike trains with any pairwise correlation can be

generated, and higher-order correlations are compatible with common synaptic input. Relations between allowable mean rates and correlations within a population are discussed. The algorithm is highly efficient, its complexity increasing linearly with the number of spike trains generated and therefore inversely with the number of cross-correlated pairs.

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115. Input-Driven Oscillations in Networks with Excitatory and Inhibitory Neurons with Dynamic Synapses

Daniele Marinazzo, Hilbert J. Kappen, Stan Gielen

https://doi.org/10.1162/neco.2007.19.7.1739 | pp. 1739-1765

Previous work has shown that networks of neurons with two coupled layers of excitatory and inhibitory neurons can reveal oscillatory activity. For example, Börgers and Kopell (2003) have shown that oscillations occur when the excitatory neurons receive a sufficiently large input. A constant drive to the excitatory neurons is sufficient for oscillatory activity. Other studies (Doiron, Chacron, Maler, Longtin, & Bastian, 2003; Doiron, Lindner, Longtin, Maler, & Bastian, 2004) have shown that networks of neurons with two coupled layers of excitatory and inhibitory neurons reveal oscillatory activity only if the excitatory neurons receive correlated input, regardless of the amount of excitatory input. In this study, we show that these apparently contradictory results can be explained by the behavior of a single model operating in different regimes of parameter space. Moreover, we show that adding dynamic synapses in the inhibitory feedback loop provides a robust network behavior over a broad range of stimulus intensities, contrary to that of previous models. A remarkable property of the introduction of dynamic synapses is that the activity of the network reveals synchronized oscillatory components in the case of correlated input, but also reflects the temporal behavior of the input signal to the excitatory neurons. This allows the network to encode both the temporal characteristics of the input and the presence of spatial correlations in the input simultaneously.

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116. Extracting Number-Selective Responses from Coherent Oscillations in a Computer Model

Jeremy A. Miller, Garrett T. Kenyon

https://doi.org/10.1162/neco.2007.19.7.1766 | pp. 1766-1797

Cortical neurons selective for numerosity may underlie an innate number sense in both animals and humans. We hypothesize that the number- selective responses of cortical neurons may in part be extracted from coherent, object-specific oscillations . Here, indirect evidence for this hypothesis is obtained by analyzing the numerosity information encoded by coherent oscillations in artificially generated spikes trains. Several experiments report that gamma-band oscillations evoked by the same object remain coherent, whereas oscillations evoked by separate objects are uncorrelated. Because the oscillations arising from separate objects would add in random phase to the total power summed across all stimulated neurons, we postulated that the total gamma activity, normalized by the number of spikes, should fall roughly as the square root of the number of objects in the scene, thereby implicitly encoding numerosity. To test the hypothesis, we examined the normalized gamma activity in multiunit spike trains, 50 to 1000 msec in duration, produced by a model feedback circuit previously shown to generate realistic coherent oscillations. In response to images containing different numbers of objects, regardless of their shape, size, or shading, the normalized gamma activity followed a square-root-of-n rule as long as the separation between objects

was sufficiently large and their relative size and contrast differences were not too great. Arrays of winner-take-all numerosity detectors, each responding to normalized gamma activity within a particular band, exhibited tuning curves consistent with behavioral data. We conclude that coherent oscillations in principle could contribute to the number-selective responses of cortical neurons, although many critical issues await experimental resolution.

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117. Selectivity and Stability via Dendritic Nonlinearity

Kenji Morita, Masato Okada, Kazuyuki Aihara

https://doi.org/10.1162/neco.2007.19.7.1798 | pp. 1798-1853

Inspired by recent studies regarding dendritic computation, we constructed a recurrent neural network model incorporating dendritic lateral inhibition. Our model consists of an input layer and a neuron layer that includes excitatory cells and an inhibitory cell; this inhibitory cell is activated by the pooled activities of all the excitatory cells, and it in turn inhibits each dendritic branch of the excitatory cells that receive excitations from the input layer. Dendritic nonlinear operation consisting of branch-specifically rectified inhibition and saturation is described by imposing nonlinear transfer functions before summation over the branches. In this model with sufficiently strong recurrent excitation, on transiently presenting a stimulus that has a high correlation with feed- forward connections of one of the excitatory cells, the corresponding cell becomes highly active, and the activity is sustained after the stimulus is turned off, whereas all the other excitatory cells continue to have low activities. But on transiently presenting a stimulus that does not have high correlations with feedforward connections of any of the excitatory cells, all the excitatory cells continue to have low activities. Interestingly, such stimulus-selective sustained response is preserved for a wide range of stimulus intensity. We derive an analytical formulation of the model in the limit where individual excitatory cells have an infinite number of dendritic branches and prove the existence of an equilibrium point corresponding to such a balanced low-level activity state as observed in the simulations, whose stability depends solely on the signal-to-noise ratio of the stimulus. We propose this model as a model of stimulus selectivity equipped with self-sustainability and intensity-invariance simultaneously, which was difficult in the conventional competitive neural networks with a similar degree of complexity in their network architecture. We discuss the biological relevance of the model in a general framework of computational neuroscience.

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118. Filtering of Spatial Bias and Noise Inputs by Spatially Structured Neural Networks

Naoki Masuda, Masato Okada, Kazuyuki Aihara

https://doi.org/10.1162/neco.2007.19.7.1854 | pp. 1854-1870

With spatially organized neural networks, we examined how bias and noise inputs with spatial structure result in different network states such as bumps, localized oscillations, global oscillations, and localized synchronous firing that may be relevant to, for example, orientation selectivity. To this end, we used networks of McCulloch-Pitts neurons, which allow theoretical predictions, and verified the obtained results with numerical simulations. Spatial inputs, no matter whether they are bias inputs or shared noise inputs, affect only firing activities with resonant spatial frequency. The component of noise that is independent for different neurons increases the linearity of the neural system and gives rise to less spatial mode mixing and less bistability of population activities.

119. Imposing Biological Constraints onto an Abstract Neocortical Attractor Network Model

Christopher Johansson, Anders Lansner

https://doi.org/10.1162/neco.2007.19.7.1871 | pp. 1871-1896

In this letter, we study an abstract model of neocortex based on its modularization into mini- and hypercolumns. We discuss a full-scale instance of this model and connect its network properties to the underlying biological properties of neurons in cortex. In particular, we discuss how the biological constraints put on the network determine the network's performance in terms of storage capacity. We show that a network instantiating the model scales well given the biologically constrained parameters on activity and connectivity, which makes this network interesting also as an engineered system. In this model, the minicolumns are grouped into hypercolumns that can be active or quiescent, and the model predicts that only a few percent of the hypercolumns should be active at any one time. With this model, we show that at least 20 to 30 pyramidal neurons should be aggregated into a minicolumn and at least 50 to 60 minicolumns should be grouped into a hypercolumn in order to achieve high storage capacity.

120. Combining Reconstruction and Discrimination with Class-Specific Sparse Coding

Stephan Hasler, Heiko Wersing, Edgar Körner

https://doi.org/10.1162/neco.2007.19.7.1897 | pp. 1897-1918

Sparse coding is an important approach for the unsupervised learning of sensory features. In this contribution, we present two new methods that extend the traditional sparse coding approach with supervised components. Our goal is to increase the suitability of the learned features for classification tasks while keeping most of their general representation capability. We analyze the effect of the new methods using visualization on artificial data and discuss the results on two object test sets with regard to the properties of the found feature representation.

121. SVDD-Based Pattern Denoising

Jooyoung Park, Daesung Kang, Jong-Ho Kim, James T. Kwok, Ivor W. Tsang

https://doi.org/10.1162/neco.2007.19.7.1919 | pp. 1919-1938

The support vector data description (SVDD) is one of the best-known one-class support vector learning methods, in which one tries the strategy of using balls defined on the feature space in order to distinguish a set of normal data from all other possible abnormal objects. The major concern of this letter is to extend the main idea of SVDD to pattern denoising. Combining the geodesic projection to the spherical decision boundary resulting from the SVDD, together with solving the preimage problem, we propose a new method for pattern denoising. We first solve SVDD for the training data and then for each noisy test pattern, obtain its denoised feature by moving its feature vector along the geodesic on the manifold to the nearest decision boundary of the SVDD ball. Finally we find the location of the denoised pattern by obtaining the pre-image of the denoised feature. The applicability of the proposed method is illustrated by a number of toy and real-world data sets.

122. Feature Selection via Coalitional Game Theory

Shay B. Cohen, Gideon Dror, Eytan Ruppin

https://doi.org/10.1162/neco.2007.19.7.1939 | pp. 1939-1961

We present and study the contribution-selection algorithm (CSA), a novel algorithm for feature selection. The algorithm is based on the multiperturbation shapley analysis (MSA), a framework that relies on game theory to estimate usefulness. The algorithm iteratively estimates the usefulness of features and selects them accordingly, using either forward selection or backward elimination. It can optimize various performance measures over unseen data such as accuracy, balanced error rate, and area under receiver-operator-characteristic curve. Empirical comparison with several other existing feature selection methods shows that the backward elimination variant of CSA leads to the most accurate classification results on an array of data sets.

123. Outliers Detection in Multivariate Time Series by Independent Component Analysis

Roberto Baragona, Francesco Battaglia

https://doi.org/10.1162/neco.2007.19.7.1962 | pp. 1962-1984

In multivariate time series, outlying data may be often observed that do not fit the common pattern. Occurrences of outliers are unpredictable events that may severely distort the analysis of the multivariate time series. For instance, model building, seasonality assessment, and forecasting may be seriously affected by undetected outliers. The structure dependence of the multivariate time series gives rise to the well-known smearing and masking phenomena that prevent using most outliers' identification techniques. It may be noticed, however, that a convenient way for representing multiple outliers consists of superimposing a deterministic disturbance to a gaussian multivariate time series. Then outliers may be modeled as nongaussian time series components. Independent component analysis is a recently developed tool that is likely to be able to extract possible outlier patterns. In practice, independent component analysis may be used to analyze multivariate observable time series and separate regular and outlying unobservable components. In the factor models framework too, it is shown that independent component analysis is a useful tool for detection of outliers in multivariate time series. Some algorithms that perform independent component analysis are compared. It has been found that all algorithms are effective in detecting various types of outliers, such as patches, level shifts, and isolated outliers, even at the beginning or the end of the stretch of observations. Also, there is no appreciable difference in the ability of different algorithms to display the outlying observations pattern.

124. Models Wagging the Dog: Are Circuits Constructed with Disparate Parameters?

Thomas Nowotny, Attila Szücs, Rafael Levi, Allen I. Selverston

https://doi.org/10.1162/neco.2007.19.8.1985 | pp. 1985-2003

In a recent article, Prinz, Bucher, and Marder (2004) addressed the fundamental question of whether neural systems are built with a fixed blueprint of tightly controlled parameters or in a way in which properties can

vary largely from one individual to another, using a database modeling approach. Here, we examine the main conclusion that neural circuits indeed are built with largely varying parameters in the light of our own experimental and modeling observations. We critically discuss the experimental and theoretical evidence, including the general adequacy of database approaches for questions of this kind, and come to the conclusion that the last word for this fundamental question has not yet been spoken.

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125. Multiplicative Updates for Nonnegative Quadratic Programming

Fei Sha, Yuanqing Lin, Lawrence K. Saul, Daniel D. Lee

https://doi.org/10.1162/neco.2007.19.8.2004 | pp. 2004-2031

Many problems in neural computation and statistical learning involve optimizations with nonnegativity constraints. In this article, we study convex problems in quadratic programming where the optimization is confined to an axis-aligned region in the nonnegative orthant. For these problems, we derive multiplicative updates that improve the value of the objective function at each iteration and converge monotonically to the global minimum. The updates have a simple closed form and do not involve any heuristics or free parameters that must be tuned to ensure convergence. Despite their simplicity, they differ strikingly in form from other multiplicative updates used in machine learning. We provide complete proofs of convergence for these updates and describe their application to problems in signal processing and pattern recognition.

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126. Critical Analysis of Dimension Reduction by a Moment Closure Method in a Population Density Approach to Neural Network Modeling

Cheng Ly, Daniel Tranchina

https://doi.org/10.1162/neco.2007.19.8.2032 | pp. 2032-2092

Computational techniques within the population density function (PDF) framework have provided time-saving alternatives to classical Monte Carlo simulations of neural network activity. Efficiency of the PDF method is lost as the underlying neuron model is made more realistic and the number of state variables increases. In a detailed theoretical and computational study, we elucidate strengths and weaknesses of dimension reduction by a particular moment closure method (Cai, Tao, Shelley, & McLaughlin, 2004; Cai, Tao, Rangan, & McLaughlin, 2006) as applied to integrate-and-fire neurons that receive excitatory synaptic input only. When the unitary postsynaptic conductance event has a single-exponential time course, the evolution equation for the PDF is a partial differential integral equation in two state variables, voltage and excitatory conductance. In the moment closure method, one approximates the conditional kth centered moment of excitatory conductance given voltage by the corresponding unconditioned moment. The result is a system of k coupled partial differential equations with one state variable, voltage, and k coupled ordinary differential equations. Moment closure at k = 2 works well, and at k = 3 works even better, in the regime of high dynamically varying synaptic input rates. Both closures break down at lower synaptic input rates. Phase-plane analysis of the k = 2 problem with typical parameters proves, and reveals why, no steady-state solutions exist below a synaptic input rate that gives a firing rate of 59 s(1) in the full 2D problem. Closure at k = 3 fails for similar reasons. Low firing-rate solutions can be obtained only with parameters for the amplitude or kinetics (or both) of the unitary postsynaptic conductance event that are on the edge of the physiological range. We conclude that this dimension-reduction method gives ill-posed problems for a wide range of physiological parameters, and we suggest future directions.

127. Oscillatory Neural Networks with Self-Organized Segmentation of Overlapping Patterns

Thomas Burwick

https://doi.org/10.1162/neco.2007.19.8.2093 | pp. 2093-2123

Temporal coding is considered with an oscillatory network model that generalizes the Cohen-Grossberg-Hopfield model. It is assumed that the frequency of oscillating units increases with stronger and more coherent input. We refer to this mechanism as acceleration. In the context of Hebbian memory, synchronization and acceleration take complementary roles, and their combined effect on the storage of patterns is profound. Acceleration implies the desynchronization that is needed for self-organized segmention of two overlapping patterns. The superposition problem is thereby solved even without including competition couplings. With respect to brain dynamics, we point to analogies with oscillation spindles in the gamma range and responses to perceptual rivalries.

128. Multistability in Spiking Neuron Models of Delayed Recurrent Inhibitory Loops

Jianfu Ma, Jianhong Wu

https://doi.org/10.1162/neco.2007.19.8.2124 | pp. 2124-2148

We consider the effect of the effective timing of a delayed feedback on the excitatory neuron in a recurrent inhibitory loop, when biological realities of firing and absolute refractory period are incorporated into a phenomenological spiking linear or quadratic integrate-and-fire neuron model. We show that such models are capable of generating a large number of asymptotically stable periodic solutions with predictable patterns of oscillations. We observe that the number of fixed points of the so-called phase resetting map coincides with the number of distinct periods of all stable periodic solutions rather than the number of stable patterns. We demonstrate how configurational information corresponding to these distinct periods can be explored to calculate and predict the number of stable patterns.

129. Analysis and Design of Associative Memories Based on Recurrent Neural Networks with Linear Saturation Activation Functions and Time-Varying Delays

Zhigang Zeng, Jun Wang

https://doi.org/10.1162/neco.2007.19.8.2149 | pp. 2149-2182

In this letter, some sufficient conditions are obtained to guarantee recurrent neural networks with linear saturation activation functions, and time-varying delays have multiequilibria located in the saturation region and the boundaries of the saturation region. These results on pattern characterization are used to analyze and design autoassociative memories, which are directly based on the parameters of the neural networks. Moreover, a formula for the numbers of spurious equilibria is also derived. Four design procedures for recurrent neural networks with linear saturation activation functions and time-varying delays are developed based on stability results. Two of these procedures allow the neural network to be capable of learning and

forgetting. Finally, simulation results demonstrate the validity and characteristics of the proposed approach.

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130. Robust Loss Functions for Boosting

Takafumi Kanamori, Takashi Takenouchi, Shinto Eguchi, Noboru Murata

https://doi.org/10.1162/neco.2007.19.8.2183 | pp. 2183-2244

Boosting is known as a gradient descent algorithm over loss functions. It is often pointed out that the typical boosting algorithm, Adaboost, is highly affected by outliers. In this letter, loss functions for robust boosting are studied. Based on the concept of robust statistics, we propose a transformation of loss functions that makes boosting algorithms robust against extreme outliers. Next, the truncation of loss functions is applied to contamination models that describe the occurrence of mislabels near decision boundaries. Numerical experiments illustrate that the proposed loss functions derived from the contamination models are useful for handling highly noisy data in comparison with other loss functions.

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131. Reinforcement Learning, Spike-Time-Dependent Plasticity, and the BCM Rule

Dorit Baras, Ron Meir

https://doi.org/10.1162/neco.2007.19.8.2245 | pp. 2245-2279

Learning agents, whether natural or artificial, must update their internal parameters in order to improve their behavior over time. In reinforcement learning, this plasticity is influenced by an environmental signal, termed a reward, that directs the changes in appropriate directions. We apply a recently introduced policy learning algorithm from machine learning to networks of spiking neurons and derive a spike-time-dependent plasticity rule that ensures convergence to a local optimum of the expected average reward. The approach is applicable to a broad class of neuronal models, including the Hodgkin-Huxley model. We demonstrate the effectiveness of the derived rule in several toy problems. Finally, through statistical analysis, we show that the synaptic plasticity rule established is closely related to the widely used BCM rule, for which good biological evidence exists.

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132. A Multichip Neuromorphic System for Spike-Based Visual Information Processing

R. Jacob Vogelstein, Udayan Mallik, Eugenio Culurciello, Gert Cauwenberghs, Ralph Etienne■Cummings

https://doi.org/10.1162/neco.2007.19.9.2281 | pp. 2281-2300

We present a multichip, mixed-signal VLSI system for spike-based vision processing. The system consists of an 80 x 60 pixel neuromorphic retina and a 4800 neuron silicon cortex with 4,194,304 synapses. Its functionality is illustrated with experimental data on multiple components of an attention-based hierarchical model of cortical object recognition, including feature coding, salience detection, and foveation. This model exploits arbitrary and reconfigurable connectivity between cells in the multichip architecture, achieved by asynchronously routing neural spike events within and between chips according to a memory-based look-up table. Synaptic parameters, including conductance and reversal potential, are also stored in memory and are used to dynamically configure synapse circuits within the silicon neurons.

133. Visual Recognition and Inference Using Dynamic Overcomplete Sparse Learning

Joseph F. Murray, Kenneth Kreutz-Delgado

https://doi.org/10.1162/neco.2007.19.9.2301 | pp. 2301-2352

We present a hierarchical architecture and learning algorithm for visual recognition and other visual inference tasks such as imagination, reconstruction of occluded images, and expectation-driven segmentation. Using properties of biological vision for guidance, we posit a stochastic generative world model and from it develop a simplified world model (SWM) based on a tractable variational approximation that is designed to enforce sparse coding. Recent developments in computational methods for learning overcomplete representations (Lewicki & Sejnowski, 2000; Teh, Welling, Osindero, & Hinton, 2003) suggest that overcompleteness can be useful for visual tasks, and we use an overcomplete dictionary learning algorithm (Kreutz-Delgado, et al., 2003) as a preprocessing stage to produce accurate, sparse codings of images. Inference is performed by constructing a dynamic multilayer network with feedforward, feedback, and lateral connections, which is trained to approximate the SWM. Learning is done with a variant of the back-propagation-through-time algorithm, which encourages convergence to desired states within a fixed number of iterations. Vision tasks require large networks, and to make learning efficient, we take advantage of the sparsity of each layer to update only a small subset of elements in a large weight matrix at each iteration. Experiments on a set of rotated objects demonstrate various types of visual inference and show that increasing the degree of overcompleteness improves recognition performance in difficult scenes with occluded objects in clutter.

134. Visual Remapping by Vector Subtraction: Analysis of Multiplicative Gain Field Models

Carlos R. Cassanello, Vincent P. Ferrera

https://doi.org/10.1162/neco.2007.19.9.2353 | pp. 2353-2386

Saccadic eye movements remain spatially accurate even when the target becomes invisible and the initial eye position is perturbed. The brain accomplishes this in part by remapping the remembered target location in retinal coordinates. The computation that underlies this visual remapping is approximated by vector subtraction: the original saccade vector is updated by subtracting the vector corresponding to the intervening eye movement. The neural mechanism by which vector subtraction is implemented is not fully understood. Here, we investigate vector subtraction within a framework in which eye position and retinal target position signals interact multiplicatively (gain field). When the eyes move, they induce a spatial modulation of the firing rates across a retinotopic map of neurons. The updated saccade metric can be read from the shift of the peak of the population activity across the map. This model uses a quasi-linear (half-rectified) dependence on the eye position and requires the slope of the eye position input to be negatively proportional to the preferred retinal position of each neuron. We derive analytically this constraint and study its range of validity. We discuss how this mechanism relates to experimental results reported in the frontal eye fields of macaque monkeys.

135. Representations of Space and Time in the Maximization of Information Flow in the Perception-Action Loop

Alexander S. Klyubin, Daniel Polani, Chrystopher L. Nehaniv

https://doi.org/10.1162/neco.2007.19.9.2387 | pp. 2387-2432

Sensor evolution in nature aims at improving the acquisition of information from the environment and is intimately related with selection pressure toward adaptivity and robustness. Our work in the area indicates that information theory can be applied to the perception-action loop. This letter studies the perception-action loop of agents, which is modeled as a causal Bayesian network. Finite state automata are evolved as agent controllers in a simple virtual world to maximize information flow through the perception-action loop. The information flow maximization organizes the agent's behavior as well as its information processing. To gain more insight into the results, the evolved implicit representations of space and time are analyzed in an information-theoretic manner, which paves the way toward a principled and general understanding of the mechanisms guiding the evolution of sensors in nature and provides insights into the design of mechanisms for artificial sensor evolution.

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136. Computational Properties of Networks of Synchronous Groups of Spiking Neurons

Judith E. Dayhoff

https://doi.org/10.1162/neco.2007.19.9.2433 | pp. 2433-2467

We demonstrate a model in which synchronously firing ensembles of neurons are networked to produce computational results. Each ensemble is a group of biological integrate-and-fire spiking neurons, with probabilistic interconnections between groups. An analogy is drawn in which each individual processing unit of an artificial neural network corresponds to a neuronal group in a biological model. The activation value of a unit in the artificial neural network corresponds to the fraction of active neurons, synchronously firing, in a biological neuronal group. Weights of the artificial neural network correspond to the product of the interconnection density between groups, the group size of the presynaptic group, and the postsynaptic potential heights in the synchronous group model. All three of these parameters can modulate connection strengths between neuronal groups in the synchronous group models. We give an example of nonlinear classification (XOR) and a function approximation example in which the capability of the artificial neural network can be captured by a neural network model with biological integrate-and-fire neurons configured as a network of synchronously firing ensembles of such neurons. We point out that the general function approximation capability proven for feedforward artificial neural networks appears to be approximated by networks of neuronal groups that fire in synchrony, where the groups comprise integrate-and-fire neurons. We discuss the advantages of this type of model for biological systems, its possible learning mechanisms, and the associated timing relationships.

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137. Variable Timescales of Repeated Spike Patterns in Synfire Chain with Mexican-Hat Connectivity

Kosuke Hamaguchi, Masato Okada, Kazuyuki Aihara

https://doi.org/10.1162/neco.2007.19.9.2468 | pp. 2468-2491

Repetitions of precise spike patterns observed both in vivo and in vitro have been reported for more than a decade. Studies on the spike volley (a pulse packet) propagating through a homogeneous feedforward network have demonstrated its capability of generating spike patterns with millisecond fidelity. This model is called the synfire chain and suggests a possible mechanism for generating repeated spike patterns (RSPs). The propagation speed of the pulse packet determines the temporal property of RSPs. However, the relationship between propagation speed and network structure is not well understood. We studied a feedforward network with Mexican-hat connectivity by using the leaky integrate-and-fire neuron model and analyzed the network dynamics with the Fokker-Planck equation. We examined the effect of the spatial pattern of pulse packets on RSPs in the network with multistability. Pulse packets can take spatially uniform or localized shapes in a multistable regime, and they propagate with different speeds. These distinct pulse packets generate RSPs with different timescales, but the order of spikes and the ratios between interspike intervals are preserved. This result indicates that the RSPs can be transformed into the same template pattern through the expanding or contracting operation of the timescale.

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138. Asymptotic Behavior and Synchronizability Characteristics of a Class of Recurrent Neural Networks

Christof Cebulla

https://doi.org/10.1162/neco.2007.19.9.2492 | pp. 2492-2514

We propose an approach to the analysis of the influence of the topology of a neural network on its synchronizability in the sense of equal output activity rates given by a particular neural network model. The model we introduce is a variation of the Zhang model. We investigate the time-asymptotic behavior of the corresponding dynamical system (in particular, the conditions for the existence of an invariant compact asymptotic set) and apply the results of the synchronizability analysis on a class of random scale free networks and to the classical random networks with Poisson connectivity distribution.

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139. Self-Organizing Maps with Asymmetric Neighborhood Function

Takaaki Aoki, Toshio Aoyagi

https://doi.org/10.1162/neco.2007.19.9.2515 | pp. 2515-2535

The self-organizing map (SOM) is an unsupervised learning method as well as a type of nonlinear principal component analysis that forms a topologically ordered mapping from the high-dimensional data space to a low-dimensional representation space. It has recently found wide applications in such areas as visualization, classification, and mining of various data. However, when the data sets to be processed are very large, a copious amount of time is often required to train the map, which seems to restrict the range of putative applications. One of the major culprits for this slow ordering time is that a kind of topological defect (e.g., a kink in one dimension or a twist in two dimensions) gets created in the map during training. Once such a defect appears in the map during training, the ordered map cannot be obtained until the defect is eliminated, for which the number of iterations required is typically several times larger than in the absence of the defect. In order to overcome this weakness, we propose that an asymmetric neighborhood function be used for the SOM algorithm. Compared with the commonly used symmetric neighborhood function, we found that an asymmetric neighborhood function accelerates the ordering process of the SOM algorithm, though this asymmetry tends to distort the generated ordered map. We demonstrate that the distortion of the map can

be suppressed by improving the asymmetric neighborhood function SOM algorithm. The number of learning steps required for perfect ordering in the case of the one-dimensional SOM is numerically shown to be reduced from O(N(3)) to O(N(2)) with an asymmetric neighborhood function, even when the improved algorithm is used to get the final map without distortion.

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140. Parametric Embedding for Class Visualization

Tomoharu Iwata, Kazumi Saito, Naonori Ueda, Sean Stromsten, Thomas L. Griffiths, Joshua B. Tenenbaum
https://doi.org/10.1162/neco.2007.19.9.2536 | pp. 2536-2556

We propose a new method, parametric embedding (PE), that embeds objects with the class structure into a low-dimensional visualization space. PE takes as input a set of class conditional probabilities for given data points and tries to preserve the structure in an embedding space by minimizing a sum of Kullback-Leibler divergences, under the assumption that samples are generated by a gaussian mixture with equal covariances in the embedding space. PE has many potential uses depending on the source of the input data, providing insight into the classifier's behavior in supervised, semisupervised, and unsupervised settings. The PE algorithm has a computational advantage over conventional embedding methods based on pairwise object relations since its complexity scales with the product of the number of objects and the number of classes. We demonstrate PE by visualizing supervised categorization of Web pages, semisupervised categorization of digits, and the relations of words and latent topics found by an unsupervised algorithm, latent Dirichlet allocation.

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141. MISEP Method for Postnonlinear Blind Source Separation

Chun-Hou Zheng, De-Shuang Huang, Kang Li, George Irwin, Zhan-Li Sun
https://doi.org/10.1162/neco.2007.19.9.2557 | pp. 2557-2578

In this letter, a standard postnonlinear blind source separation algorithm is proposed, based on the MISEP method, which is widely used in linear and nonlinear independent component analysis. To best suit a wide class of postnonlinear mixtures, we adapt the MISEP method to incorporate a priori information of the mixtures. In particular, a group of three-layered perceptrons and a linear network are used as the unmixing system to separate sources in the postnonlinear mixtures, and another group of three-layered perceptron is used as the auxiliary network. The learning algorithm for the unmixing system is then obtained by maximizing the output entropy of the auxiliary network. The proposed method is applied to postnonlinear blind source separation of both simulation signals and real speech signals, and the experimental results demonstrate its effectiveness and efficiency in comparison with existing methods.

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142. Erratum

(authors unknown)
https://doi.org/10.1162/neco.2007.19.9.2579 | pp. 2579-2579

No abstract available.

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143. Synaptic Dynamics in Analog VLSI

Chiara Bartolozzi, Giacomo Indiveri

https://doi.org/10.1162/neco.2007.19.10.2581 | pp. 2581-2603

Synapses are crucial elements for computation and information transfer in both real and artificial neural systems. Recent experimental findings and theoretical models of pulse-based neural networks suggest that synaptic dynamics can play a crucial role for learning neural codes and encoding spatiotemporal spike patterns. Within the context of hardware implementations of pulse-based neural networks, several analog VLSI circuits modeling synaptic functionality have been proposed. We present an overview of previously proposed circuits and describe a novel analog VLSI synaptic circuit suitable for integration in large VLSI spike-based neural systems. The circuit proposed is based on a computational model that fits the real postsynaptic currents with exponentials. We present experimental data showing how the circuit exhibits realistic dynamics and show how it can be connected to additional modules for implementing a wide range of synaptic properties.

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144. Exact Simulation of Integrate-and-Fire Models with Exponential Currents

Romain Brette

https://doi.org/10.1162/neco.2007.19.10.2604 | pp. 2604-2609

Neural networks can be simulated exactly using event-driven strategies, in which the algorithm advances directly from one spike to the next spike. It applies to neuron models for which we have (1) an explicit expression for the evolution of the state variables between spikes and (2) an explicit test on the state variables that predicts whether and when a spike will be emitted. In a previous work, we proposed a method that allows exact simulation of an integrate-and-fire model with exponential conductances, with the constraint of a single synaptic time constant. In this note, we propose a method, based on polynomial root finding, that applies to integrate-and-fire models with exponential currents, with possibly many different synaptic time constants. Models can include biexponential synaptic currents and spike-triggered adaptation currents.

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145. The Relation Between Color Discrimination and Color Constancy: When Is Optimal Adaptation Task Dependent?

Alicia B. Abrams, James Hillis, David H. Brainard

https://doi.org/10.1162/neco.2007.19.10.2610 | pp. 2610-2637

Color vision supports two distinct visual functions: discrimination and constancy. Discrimination requires that the visual response to distinct objects within a scene be different. Constancy requires that the visual response to any object be the same across scenes. Across changes in scene, adaptation can improve discrimination by optimizing the use of the available response range. Similarly, adaptation can improve constancy by stabilizing the visual response to any fixed object across changes in illumination. Can common mechanisms of adaptation achieve these two goals simultaneously? We develop a theoretical framework for answering this question and present several example calculations. In the examples studied, the answer is largely yes when the change of scene consists of a change in illumination and considerably less so when the change of scene consists of a change in the statistical ensemble of surface reflectances in the

environment.

146. Deviation from Weber's Law in the Non-Pacinian I Tactile Channel: A Psychophysical and Simulation Study of Intensity Discrimination

Burak Güçlü

https://doi.org/10.1162/neco.2007.19.10.2638 | pp. 2638-2664

This study involves psychophysical experiments and computer simulations to investigate intensity discrimination in the non-Pacinian I (NP I) tactile channel. The simulations were based on an established population model for rapidly adapting mechanoreceptive fibers (Güuçlü & Bolanowski, 2004a). Several intensity codes were tested as decision criteria: number of active neurons, total spike count, maximal spike count, distribution of spike counts among the afferent population, and synchronization of spike times. Simulations that used the number of active fibers as the intensity code gave the most accurate results. However, the Weber fractions obtained from simulations are smaller than psychophysical Weber fractions, which suggests that only a subset of the afferent population is recruited for intensity discrimination during psychophysical experiments. Simulations could also capture the deviation from Weber's law, that is, the decrease of the Weber fraction as a function of the stimulus level, which was present in the psychophysical data. Since the psychophysical task selectively activated the NP I channel, the deviation effect is probably not due to the contribution of another tactile channel but rather is explicitly produced by the NP I channel. Moreover, because simulations with all tested intensity codes resulted in the same effect, the activity of the afferent population is sufficient to explain the deviation, without the need for a higher-order network. Depending on the intensity code used, the mechanical spread of the stimulus, rate-intensity functions of the tactile fibers, and the decreasing spike-phase jitter contribute to the deviation from Weber's law.

147. Learning the Lie Groups of Visual Invariance

Miao Xu, Rajesh P. N. Rao

https://doi.org/10.1162/neco.2007.19.10.2665 | pp. 2665-2693

A fundamental problem in biological and machine vision is visual invariance: How are objects perceived to be the same despite transformations such as translations, rotations, and scaling? In this letter, we describe a new, unsupervised approach to learning invariances based on Lie group theory. Unlike traditional approaches that sacrifice information about transformations to achieve invariance, the Lie group approach explicitly models the effects of transformations in images. As a result, estimates of transformations are available for other purposes, such as pose estimation and visuomotor control. Previous approaches based on first-order Taylor series expansions of images can be regarded as special cases of the Lie group approach, which utilizes a matrix-exponential-based generative model of images and can handle arbitrarily large transformations. We present an unsupervised expectation-maximization algorithm for learning Lie transformation operators directly from image data containing examples of transformations. Our experimental results show that the Lie operators learned by the algorithm from an artificial data set containing six types of affine transformations closely match the analytically predicted affine operators. We then demonstrate that the algorithm can also recover novel transformation operators from natural image sequences. We conclude by showing that the learned operators can be used to both generate and estimate transformations in images, thereby providing a basis for achieving visual invariance.

148. Learning with “Relevance”: Using a Third Factor to Stabilize Hebbian Learning

Bernd Porr, Florentin Wörgötter

https://doi.org/10.1162/neco.2007.19.10.2694 | pp. 2694-2719

It is a well-known fact that Hebbian learning is inherently unstable because of its self-amplifying terms: the more a synapse grows, the stronger the postsynaptic activity, and therefore the faster the synaptic growth. This unwanted weight growth is driven by the autocorrelation term of Hebbian learning where the same synapse drives its own growth. On the other hand, the cross-correlation term performs actual learning where different inputs are correlated with each other. Consequently, we would like to minimize the autocorrelation and maximize the cross-correlation. Here we show that we can achieve this with a third factor that switches on learning when the autocorrelation is minimal or zero and the cross-correlation is maximal. The biological counterpart of such a third factor is a neuromodulator that switches on learning at a certain moment in time. We show in a behavioral experiment that our three-factor learning clearly outperforms classical Hebbian learning.

149. Synchrony-Induced Switching Behavior of Spike Pattern Attractors Created by Spike-Timing-Dependent Plasticity

Takaaki Aoki, Toshio Aoyagi

https://doi.org/10.1162/neco.2007.19.10.2720 | pp. 2720-2738

Although context-dependent spike synchronization among populations of neurons has been experimentally observed, its functional role remains controversial. In this modeling study, we demonstrate that in a network of spiking neurons organized according to spike-timing-dependent plasticity, an increase in the degree of synchrony of a uniform input can cause transitions between memorized activity patterns in the order presented during learning. Furthermore, context-dependent transitions from a single pattern to multiple patterns can be induced under appropriate learning conditions. These findings suggest one possible functional role of neuronal synchrony in controlling the flow of information by altering the dynamics of the network.

150. Competition Between Synaptic Depression and Facilitation in Attractor Neural Networks

Joaqu■■n J. Torres, Jesús M. Cortés, Joaquín Marro, Hilbert J. Kappen

https://doi.org/10.1162/neco.2007.19.10.2739 | pp. 2739-2755

We study the effect of competition between short-term synaptic depression and facilitation on the dynamic properties of attractor neural networks, using Monte Carlo simulation and a mean-field analysis. Depending on the balance of depression, facilitation, and the underlying noise, the network displays different behaviors, including associative memory and switching of activity between different attractors. We conclude that synaptic facilitation enhances the attractor instability in a way that (1) intensifies the system adaptability to external stimuli, which is in agreement with experiments, and (2) favors the retrieval of information with less error during short time intervals.

151. Projected Gradient Methods for Nonnegative Matrix Factorization

Chih■Jen Lin

https://doi.org/10.1162/neco.2007.19.10.2756 | pp. 2756-2779

Nonnegative matrix factorization (NMF) can be formulated as a minimization problem with bound constraints. Although bound-constrained optimization has been studied extensively in both theory and practice, so far no study has formally applied its techniques to NMF. In this letter, we propose two projected gradient methods for NMF, both of which exhibit strong optimization properties. We discuss efficient implementations and demonstrate that one of the proposed methods converges faster than the popular multiplicative update approach. A simple Matlab code is also provided.

152. Integration of Stochastic Models by Minimizing α-Divergence

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https://doi.org/10.1162/neco.2007.19.10.2780 | pp. 2780-2796

When there are a number of stochastic models in the form of probability distributions, one needs to integrate them. Mixtures of distributions are frequently used, but exponential mixtures also provide a good means of integration. This letter proposes a one-parameter family of integration, called alpha-integration, which includes all of these well-known integrations. These are generalizations of various averages of numbers such as arithmetic, geometric, and harmonic averages. There are psychophysical experiments that suggest that alpha-integrations are used in the brain. The alpha-divergence between two distributions is defined, which is a natural generalization of Kullback-Leibler divergence and Hellinger distance, and it is proved that alpha-integration is optimal in the sense of minimizing alpha-divergence. The theory is applied to generalize the mixture of experts and the product of experts to the alpha-mixture of experts. The alpha-predictive distribution is also stated in the Bayesian framework.

153. Distortion of Neural Signals by Spike Coding

David H. Goldberg, Andreas G. Andreou

https://doi.org/10.1162/neco.2007.19.10.2797 | pp. 2797-2839

Analog neural signals must be converted into spike trains for transmission over electrically leaky axons. This spike encoding and subsequent decoding leads to distortion. We quantify this distortion by deriving approximate expressions for the mean square error between the inputs and outputs of a spiking link. We use integrate-and-fire and Poisson encoders to convert naturalistic stimuli into spike trains and spike count and inter-spike interval decoders to generate reconstructions of the stimulus. The distortion expressions enable us to compare these spike coding schemes over a large parameter space. We verify that the integrate-and-fire encoder is more effective than the Poisson encoder. The disparity between the two encoders diminishes as the stimulus coefficient of variation (CV) increases, at which point, the variability attributed to the stimulus overwhelms the variability attributed to Poisson statistics. When the stimulus CV is small, the interspike interval decoder is superior, as the distortion resulting from spike count decoding is dominated by a term that is attributed to the discrete nature of the spike count. In this regime, additive noise has a greater impact on the interspike interval decoder than the spike count decoder. When the stimulus CV

is large, the average signal excursion is much larger than the quantization step size, and spike count decoding is superior.

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154. Gap-Based Estimation: Choosing the Smoothing Parameters for Probabilistic and General Regression Neural Networks

Mingyu Zhong, D. Coggeshall, E. Ghaneie, Thomas Pope, Mark Rivera, Michael Georgiopoulos, Georgios C. Anagnostopoulos, Mansooreh Mollaghasemi, S.M. Richie

https://doi.org/10.1162/neco.2007.19.10.2840 | pp. 2840-2864

Probabilistic neural networks (PNN) and general regression neural networks (GRNN) represent knowledge by simple but interpretable models that approximate the optimal classifier or predictor in the sense of expected value of the accuracy. These models require the specification of an important smoothing parameter, which is usually chosen by cross-validation or clustering. In this article, we demonstrate the problems with the cross-validation and clustering approaches to specify the smoothing parameter, discuss the relationship between this parameter and some of the data statistics, and attempt to develop a fast approach to determine the optimal value of this parameter. Finally, through experimentation, we show that our approach, referred to as a gap-based estimation approach, is superior in speed to the compared approaches, including support vector machine, and yields good and stable accuracy.

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155. J4 at Sweet 16: A New Wrinkle?

Bardia F. Behabadi, Bartlett W. Mel

https://doi.org/10.1162/neco.2007.19.11.2865 | pp. 2865-2870

Compartmental models provide a major source of insight into the information processing functions of single neurons. Over the past 15 years, one of the most widely used neuronal morphologies has been the cell called "j4," a layer 5 pyramidal cell from cat visual cortex originally described in Douglas, Martin, and Whitteridge (1991). The cell has since appeared in at least 28 published compartmental modeling studies, including several in this journal. In recently examining why we could not reproduce certain in vitro data involving the attenuation of signals originating in distal basal dendrites, we discovered that pronounced fluctuations in the diameter measurements of j4 lead to a bottlenecking effect that increases distal input resistances and significantly reduces voltage transfer between distal sites and the cell body. Upon smoothing these diameter fluctuations, bringing j4 more in line with other reconstructions of layer 5 pyramidal neurons, we found that the attenuation of steady-state voltage signals traveling to the cell body V(distal)/V(soma) was reduced by 60% at some locations in some branches (corresponding to a 2.5-fold increase in the voltage response at the soma for the same distal depolarization) and by 30% on average (corresponding to a 45% increase in somatic response). Changes of this magnitude could lead to different outcomes in some types of compartmental modeling studies. A smoothed version of the j4 morphology is available online at http://lnc.usc.edu/j4-smooth/ .

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156. On the Consistency of Bayesian Function Approximation Using Step Functions

Heng Lian

https://doi.org/10.1162/neco.2007.19.11.2871 | pp. 2871-2880

We consider the problem of estimating a step function with an unknown number of jumps under noisy observations on a grid. Under mild assumptions, the Bayesian approach is shown to produce a consistent estimate, even when the underlying true function is not piecewise constant. A simple prior is constructed to illustrate our assumptions.

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157. Learning Real-World Stimuli in a Neural Network with Spike-Driven Synaptic Dynamics

J. M. Brader, Walter Senn, Stefano Fusi

https://doi.org/10.1162/neco.2007.19.11.2881 | pp. 2881-2912

We present a model of spike-driven synaptic plasticity inspired by experimental observations and motivated by the desire to build an electronic hardware device that can learn to classify complex stimuli in a semisupervised fashion. During training, patterns of activity are sequentially imposed on the input neurons, and an additional instructor signal drives the output neurons toward the desired activity. The network is made of integrate-and-fire neurons with constant leak and a floor. The synapses are bistable, and they are modified by the arrival of presynaptic spikes. The sign of the change is determined by both the depolarization and the state of a variable that integrates the postsynaptic action potentials. Following the training phase, the instructor signal is removed, and the output neurons are driven purely by the activity of the input neurons weighted by the plastic synapses. In the absence of stimulation, the synapses preserve their internal state indefinitely. Memories are also very robust to the disruptive action of spontaneous activity. A network of 2000 input neurons is shown to be able to classify correctly a large number (thousands) of highly overlapping patterns (300 classes of preprocessed Latex characters, 30 patterns per class, and a subset of the NIST characters data set) and to generalize with performances that are better than or comparable to those of artificial neural networks. Finally we show that the synaptic dynamics is compatible with many of the experimental observations on the induction of long-term modifications (spike-timing-dependent plasticity and its dependence on both the postsynaptic depolarization and the frequency of pre- and postsynaptic neurons).

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158. Tight Data-Robust Bounds to Mutual Information Combining Shuffling and Model Selection Techniques

Marcelo A. Montemurro, R Senatore, Stefano Panzeri

https://doi.org/10.1162/neco.2007.19.11.2913 | pp. 2913-2957

The estimation of the information carried by spike times is crucial for a quantitative understanding of brain function, but it is difficult because of an upward bias due to limited experimental sampling. We present new progress, based on two basic insights, on reducing the bias problem. First, we show that by means of a careful application of data-shuffling techniques, it is possible to cancel almost entirely the bias of the noise entropy, the most biased part of information. This procedure provides a new information estimator that is much less biased than the standard direct one and has similar variance. Second, we use a nonparametric test to determine whether all the information encoded by the spike train can be decoded assuming a

low-dimensional response model. If this is the case, the complexity of response space can be fully captured by a small number of easily sampled parameters. Combining these two different procedures, we obtain a new class of precise estimators of information quantities, which can provide data-robust upper and lower bounds to the mutual information. These bounds are tight even when the number of trials per stimulus available is one order of magnitude smaller than the number of possible responses. The effectiveness and the usefulness of the methods are tested through applications to simulated data and recordings from somatosensory cortex. This application shows that even in the presence of strong correlations, our methods constrain precisely the amount of information encoded by real spike trains recorded in vivo.

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159. Spike-Frequency Adapting Neural Ensembles: Beyond Mean Adaptation and Renewal Theories

Eilif Müller, Lars Buesing, Johannes Schemmel, Karlheinz Meier

https://doi.org/10.1162/neco.2007.19.11.2958 | pp. 2958-3010

We propose a Markov process model for spike-frequency adapting neural ensembles that synthesizes existing mean-adaptation approaches, population density methods, and inhomogeneous renewal theory, resulting in a unified and tractable framework that goes beyond renewal and mean-adaptation theories by accounting for correlations between subsequent interspike intervals. A method for efficiently generating inhomogeneous realizations of the proposed Markov process is given, numerical methods for solving the population equation are presented, and an expression for the first-order interspike interval correlation is derived. Further, we show that the full five-dimensional master equation for a conductance-based integrate-and-fire neuron with spike-frequency adaptation and a relative refractory mechanism driven by Poisson spike trains can be reduced to a two-dimensional generalization of the proposed Markov process by an adiabatic elimination of fast variables. For static and dynamic stimulation, negative serial interspike interval correlations and transient population responses, respectively, of Monte Carlo simulations of the full five-dimensional system can be accurately described by the proposed two-dimensional Markov process.

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160. Phase Transition and Hysteresis in an Ensemble of Stochastic Spiking Neurons

Andreas Kaltenbrunner, Vicenç Gómez, Vicente López

https://doi.org/10.1162/neco.2007.19.11.3011 | pp. 3011-3050

An ensemble of stochastic nonleaky integrate-and-fire neurons with global, delayed, and excitatory coupling and a small refractory period is analyzed. Simulations with adiabatic changes of the coupling strength indicate the presence of a phase transition accompanied by a hysteresis around a critical coupling strength. Below the critical coupling production of spikes in the ensemble is governed by the stochastic dynamics, whereas for coupling greater than the critical value, the stochastic dynamics loses its influence and the units organize into several clusters with self-sustained activity. All units within one cluster spike in unison, and the clusters themselves are phase-locked. Theoretical analysis leads to upper and lower bounds for the average interspike interval of the ensemble valid for all possible coupling strengths. The bounds allow calculating the limit behavior for large ensembles and characterize the phase transition analytically. These results may be extensible to pulse-coupled oscillators.

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161. Model-Based Reinforcement Learning for Partially Observable Games with Sampling-Based State Estimation

Hajime Fujita, Shin Ishii

https://doi.org/10.1162/neco.2007.19.11.3051 | pp. 3051-3087

Games constitute a challenging domain of reinforcement learning (RL) for acquiring strategies because many of them include multiple players and many unobservable variables in a large state space. The difficulty of solving such realistic multiagent problems with partial observability arises mainly from the fact that the computational cost for the estimation and prediction in the whole state space, including unobservable variables, is too heavy. To overcome this intractability and enable an agent to learn in an unknown environment, an effective approximation method is required with explicit learning of the environmental model. We present a model-based RL scheme for large-scale multiagent problems with partial observability and apply it to a card game, hearts. This game is a well-defined example of an imperfect information game and can be approximately formulated as a partially observable Markov decision process (POMDP) for a single learning agent. To reduce the computational cost, we use a sampling technique in which the heavy integration required for the estimation and prediction can be approximated by a plausible number of samples. Computer simulation results show that our method is effective in solving such a difficult, partially observable multiagent problem.

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162. Clustering Based on Gaussian Processes

Hyun-Chul Kim, Jaewook Lee

https://doi.org/10.1162/neco.2007.19.11.3088 | pp. 3088-3107

In this letter, we develop a gaussian process model for clustering. The variances of predictive values in gaussian processes learned from a training data are shown to comprise an estimate of the support of a probability density function. The constructed variance function is then applied to construct a set of contours that enclose the data points, which correspond to cluster boundaries. To perform clustering tasks of the data points, an associated dynamical system is built, and its topological invariant property is investigated. The experimental results show that the proposed method works successfully for clustering problems with arbitrary shapes.

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163. Elman Backpropagation as Reinforcement for Simple Recurrent Networks

André Grüning

https://doi.org/10.1162/neco.2007.19.11.3108 | pp. 3108-3131

Simple recurrent networks (SRNs) in symbolic time-series prediction (e.g., language processing models) are frequently trained with gradient descent--based learning algorithms, notably with variants of backpropagation (BP). A major drawback for the cognitive plausibility of BP is that it is a supervised scheme in which a teacher has to provide a fully specified target answer. Yet agents in natural environments often receive summary feedback about the degree of success or failure only, a view adopted in reinforcement learning schemes. In this work, we show that for SRNs in prediction tasks for which there is a probability interpretation of the network's output vector, Elman BP can be reimplemented as a reinforcement learning

scheme for which the expected weight updates agree with the ones from traditional Elman BP. Network simulations on formal languages corroborate this result and show that the learning behaviors of Elman backpropagation and its reinforcement variant are very similar also in online learning tasks.

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164. Single Neuron Computation: From Dynamical System to Feature Detector

Sungho Hong, Blaise Agüera y Arcas, Adrienne L. Fairhall

https://doi.org/10.1162/neco.2007.19.12.3133 | pp. 3133-3172

White noise methods are a powerful tool for characterizing the computation performed by neural systems. These methods allow one to identify the feature or features that a neural system extracts from a complex input and to determine how these features are combined to drive the system's spiking response. These methods have also been applied to characterize the input-output relations of single neurons driven by synaptic inputs, simulated by direct current injection. To interpret the results of white noise analysis of single neurons, we would like to understand how the obtained feature space of a single neuron maps onto the biophysical properties of the membrane, in particular, the dynamics of ion channels. Here, through analysis of a simple dynamical model neuron, we draw explicit connections between the output of a white noise analysis and the underlying dynamical system. We find that under certain assumptions, the form of the relevant features is well defined by the parameters of the dynamical system. Further, we show that under some conditions, the feature space is spanned by the spike-triggered average and its successive order time derivatives.

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165. Context Learning in the Rodent Hippocampus

Mark C. Fuhs, David S. Touretzky

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We present a Bayesian statistical theory of context learning in the rodent hippocampus. While context is often defined in an experimental setting in relation to specific background cues or task demands, we advance a single, more general notion of context that suffices for a variety of learning phenomena. Specifically, a context is defined as a statistically stationary distribution of experiences, and context learning is defined as the problem of how to form contexts out of groups of experiences that cluster together in time. The challenge of context learning is solving the model selection problem: How many contexts make up the rodent's world? Solving this problem requires balancing two opposing goals: minimize the variability of the distribution of experiences within a context and minimize the likelihood of transitioning between contexts. The theory provides an understanding of why hippocampal place cell remapping sometimes develops gradually over many days of experience and why even consistent landmark differences may need to be relearned after other environmental changes. The theory provides an explanation for progressive performance improvements in serial reversal learning, based on a clear dissociation between the incremental process of context learning and the relatively abrupt context selection process. The impact of partial reinforcement on reversal learning is also addressed. Finally, the theory explains why alternating sequence learning does not consistently result in unique context-dependent sequence representations in hippocampus.

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166. Solution Methods for a New Class of Simple Model Neurons

Mark D. Humphries, Kevin Gurney

https://doi.org/10.1162/neco.2007.19.12.3216 | pp. 3216-3225

Izhikevich (2003) proposed a new canonical neuron model of spike generation. The model was surprisingly simple yet able to accurately replicate the firing patterns of different types of cortical cell. Here, we derive a solution method that allows efficient simulation of the model.

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167. Event-Driven Simulations of Nonlinear Integrate-and-Fire Neurons

Arnaud Tonnelier, Hana Belmabrouk, Dominique Martinez

https://doi.org/10.1162/neco.2007.19.12.3226 | pp. 3226-3238

Event-driven strategies have been used to simulate spiking neural networks exactly. Previous work is limited to linear integrate-and-fire neurons. In this note, we extend event-driven schemes to a class of nonlinear integrate-and-fire models. Results are presented for the quadratic integrate-and-fire model with instantaneous or exponential synaptic currents. Extensions to conductance-based currents and exponential integrate-and-fire neurons are discussed.

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168. Temporal Coding of Time-Varying Stimuli

Maoz Shamir, Kamal Sen, H. Steven Colburn

https://doi.org/10.1162/neco.2007.19.12.3239 | pp. 3239-3261

Temporal structure is an inherent property of various sensory inputs and motor outputs of the brain. For example, auditory stimuli are defined by the sound waveform. Temporal structure is also an important feature of certain visual stimuli, for example, the image on the retina of a fly during flight. In many cases, this temporal structure of the stimulus is being represented by a time-dependent neuronal activity that is locked to certain features of the stimulus. Here, we study the information capacity of the temporal code. In particular we are interested in the following questions. First, how does the information content of the code depend on the observation time of the cell's response, and what is the effect of temporal noise correlations on this information capacity? Second, what is the effect on the information content of reading the code with a finite temporal resolution for the neural response? We address these questions in the framework of a statistical model for the neuronal temporal response to a time-varying stimulus in a two-alternative forced-choice paradigm. We show that information content of the temporal response scales linearly with the overall time of the response, even in the presence of temporal noise correlations. More precisely, we find that positive temporal noise correlations have a scaling effect that decreases the information content. Nevertheless, the information content of the response continues to scale linearly with the observation time. We further show that finite temporal resolution is sufficient for obtaining most of the information from the cell's response. This finite timescale is related to the response properties of the cell.

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169. Stochastic Dynamics of a Finite-Size Spiking Neural Network

Hédi Soula, Carson C. Chow

https://doi.org/10.1162/neco.2007.19.12.3262 | pp. 3262-3292

We present a simple Markov model of spiking neural dynamics that can be analytically solved to characterize the stochastic dynamics of a finite-size spiking neural network. We give closed-form estimates for the equilibrium distribution, mean rate, variance, and autocorrelation function of the network activity. The model is applicable to any network where the probability of firing of a neuron in the network depends on only the number of neurons that fired in a previous temporal epoch. Networks with statistically homogeneous connectivity and membrane and synaptic time constants that are not excessively long could satisfy these conditions. Our model completely accounts for the size of the network and correlations in the firing activity. It also allows us to examine how the network dynamics can deviate from mean field theory. We show that the model and solutions are applicable to spiking neural networks in biophysically plausible parameter regimes.

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170. What Is the Optimal Architecture for Visual Information Routing?

Philipp Wolfrum, Christoph von der Malsburg

https://doi.org/10.1162/neco.2007.19.12.3293 | pp. 3293-3309

Analyzing the design of networks for visual information routing is an underconstrained problem due to insufficient anatomical and physiological data. We propose here optimality criteria for the design of routing networks. For a very general architecture, we derive the number of routing layers and the fanout that minimize the required neural circuitry. The optimal fanout l is independent of network size, while the number k of layers scales logarithmically (with a prefactor below 1), with the number n of visual resolution units to be routed independently. The results are found to agree with data of the primate visual system.

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171. Enhanced Sound Perception by Widespread-Onset Neuronal Responses in Auditory Cortex

Osamu Hoshino

https://doi.org/10.1162/neco.2007.19.12.3310 | pp. 3310-3334

Accumulating evidence suggests that auditory cortical neurons exhibit widespread-onset responses and restricted sustained responses to sound stimuli. When a sound stimulus is presented to a subject, the auditory cortex first responds with transient discharges across a relatively large population of neurons, showing widespread-onset responses. As time passes, the activation becomes restricted to a small population of neurons that are preferentially driven by the stimulus, showing restricted sustained responses. The sustained responses are considered to have a role in expressing information about the stimulus, but it remains to be seen what roles the widespread-onset responses have in auditory information processing. We carried out numerical simulations of a neural network model for a lateral belt area of auditory cortex. In the network, dynamic cell assemblies expressed information about auditory sounds. Lateral excitatory and inhibitory connections were made between cell assemblies, respectively, by direct and indirect projections via interneurons. Widespread-onset neuronal responses to sound stimuli (bandpassed noises) took place over the network if lateral excitation preceded lateral inhibition, making a time widow for the onset responses. The widespread-onset responses contributed to the accelerating reaction time of neurons to sensory stimulation. Lateral interaction among dynamic cell assemblies was essential for maintaining ongoing membrane potentials near thresholds for action potential generation, thereby accelerating reaction time to subsequent sensory input as well. We suggest that the widespread-onset neuronal responses and the ongoing subthreshold cortical state, for which the coordination of lateral synaptic interaction among

dissimilar cell assemblies is essential, may work together in order for the auditory cortex to quickly detect the sudden occurrence of sounds from the external environment.

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172. Bayesian Inference Explains Perception of Unity and Ventriloquism Aftereffect: Identification of Common Sources of Audiovisual Stimuli

Yoshiyuki Sat■, Taro Toyoizumi, Kazuyuki Aihara

https://doi.org/10.1162/neco.2007.19.12.3335 | pp. 3335-3355

We study a computational model of audiovisual integration by setting a Bayesian observer that localizes visual and auditory stimuli without presuming the binding of audiovisual information. The observer adopts the maximum a posteriori approach to estimate the physically delivered position or timing of presented stimuli, simultaneously judging whether they are from the same source or not. Several experimental results on the perception of spatial unity and the ventriloquism effect can be explained comprehensively if the subjects in the experiments are regarded as Bayesian observers who try to accurately locate the stimulus. Moreover, by adaptively changing the inner representation of the Bayesian observer in terms of experience, we show that our model reproduces perceived spatial frame shifts due to the audiovisual adaptation known as the ventriloquism aftereffect.

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173. Training Pi-Sigma Network by Online Gradient Algorithm with Penalty for Small Weight Update

Yan Xiong, Wei Wu, Xidai Kang, Chao Zhang

https://doi.org/10.1162/neco.2007.19.12.3356 | pp. 3356-3368

A pi-sigma network is a class of feedforward neural networks with product units in the output layer. An online gradient algorithm is the simplest and most often used training method for feedforward neural networks. But there arises a problem when the online gradient algorithm is used for pi-sigma networks in that the update increment of the weights may become very small, especially early in training, resulting in a very slow convergence. To overcome this difficulty, we introduce an adaptive penalty term into the error function, so as to increase the magnitude of the update increment of the weights when it is too small. This strategy brings about faster convergence as shown by the numerical experiments carried out in this letter.

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174. Multiorder Neurons for Evolutionary Higher-Order Clustering and Growth

Kiruthika Ramanathan, Sheng-Uei Guan

https://doi.org/10.1162/neco.2007.19.12.3369 | pp. 3369-3391

This letter proposes to use multiorder neurons for clustering irregularly shaped data arrangements. Multiorder neurons are an evolutionary extension of the use of higher-order neurons in clustering. Higher-order neurons parametrically model complex neuron shapes by replacing the classic synaptic weight by higher-order tensors. The multiorder neuron goes one step further and eliminates two problems associated with higher-order neurons. First, it uses evolutionary algorithms to select the best neuron order for a given problem. Second, it obtains more information about the underlying data distribution by identifying

the correct order for a given cluster of patterns. Empirically we observed that when the correlation of clusters found with ground truth information is used in measuring clustering accuracy, the proposed evolutionary multiorder neurons method can be shown to outperform other related clustering methods. The simulation results from the Iris, Wine, and Glass data sets show significant improvement when compared to the results obtained using self-organizing maps and higher-order neurons. The letter also proposes an intuitive model by which multiorder neurons can be grown, thereby determining the number of clusters in data.

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175. Multiple Almost Periodic Solutions in Nonautonomous Delayed Neural Networks

Kuang-Hui Lin, Chih-Wen Shih

https://doi.org/10.1162/neco.2007.19.12.3392 | pp. 3392-3420

A general methodology that involves geometric configuration of the network structure for studying multistability and multiperiodicity is developed. We consider a general class of nonautonomous neural networks with delays and various activation functions. A geometrical formulation that leads to a decomposition of the phase space into invariant regions is employed. We further derive criteria under which the n-neuron network admits 2n exponentially stable sets. In addition, we establish the existence of 2n exponentially stable almost periodic solutions for the system, when the connection strengths, time lags, and external bias are almost periodic functions of time, through applying the contraction mapping principle. Finally, three numerical simulations are presented to illustrate our theory.

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176. Index

(authors unknown)

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